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From virality to pedagogy: a systematic and comprehensive review of TikTok's educational use in South America

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TikTok's rise invites scrutiny of its educational value, yet evidence from South America remains fragmented. This study synthesizes South-American research on TikTok-mediated learning to (i) map thematic and methodological trends and (ii) quantify reported learning gains. A PRISMA-guided search of Scopus, Web of Science, SciELO and Latindex (2018–2025) yielded 1 695 records; after screening, 26 peer-reviewed studies met the inclusion criteria. Narrative synthesis traced conceptual patterns, while a meta-analytical sub-study aggregated effect data from the 15 papers that reported quantitative outcomes. Research clusters around four application domains: language learning, science/health communication, reading & digital storytelling, and civic-environmental education. Most interventions occur in secondary and higher-education settings and employ challenge-based or duet/stitch tasks that leverage TikTok's multimodal affordances for motivation and identity work. Meta-analytic pooling—limited by under-reporting of descriptive statistics—nevertheless identified large pre/post differences associated with speaking accuracy ($d = 2.13$) and moderate pre/post differences related to emotional-intelligence clarity (Wilcoxon $p < 0.05$). Qualitative evidence corroborates heightened engagement, peer collaboration and critical-media literacy, but also flags persistent digital divides, algorithmic bias and scant research in primary, indigenous and community contexts. Heterogeneous designs and incomplete statistical reporting re-strict generalizability. Future studies should adopt longitudinal, multi-site experiments and open-data practices to enable robust meta-analysis. Effective use of TikTok hinges on intentional instructional design: clear learning objectives, equity centered access strategies and critical-digital-literacy scaffolds that mitigate platform bias. This is the first systematic review and meta-analysis focused exclusively on TikTok's educational applications in South America. It provides an evidence-based agenda for researchers and practitioners seeking to harness platform-native cultures while addressing the region's sociotechnical inequalities.

KEYWORDS

digital education, educational communication, EduTok, South America, TikTok

1 Introduction

In recent years, TikTok has increasingly been used as a space of pedagogical experimentation, particularly among youth populations and educators seeking innovative approaches to digital engagement (Almoqbel et al., 2025). With over one billion monthly active users globally and an exceptionally high adoption rate among individuals aged

13–24, TikTok's influence appears to extend beyond leisure, permeating educational practices (Zhang et al., 2025). In South America, specifically South America, this influence has intersected with a complex landscape of post-pandemic educational recovery, digital acceleration, and increased attention to media-based and student-centered learning strategies (Erazo et al., 2025; Vintimilla Luzuriaga and Erazo Álvarez, 2025; Remesal and Villarroel, 2023).

However, while the global educational discourse has begun to explore TikTok's affordances in fields like language learning, digital storytelling, and critical media literacy (Zulli and Zulli, 2022), there remains a notable scarcity of integrated, regional reviews that analyze how TikTok is being used in South American educational contexts. Much of the existing literature either focuses on the platform's impact in North America and Asia or examines youth participation in general without systematically exploring formal and non-formal educational applications within South America's culturally and socioeconomically diverse territories.

This article addresses this gap by providing a comprehensive systematic and thematic review of peer-reviewed research on TikTok's use in education across South America, published between 2018 and 2025. It situates the platform within a broader theoretical and methodological framework that draws upon educational communication, media literacy, and informal-formal learning paradigms.

1.1 Theoretical foundations

1.1.1 Educational communication and participatory pedagogies

Across global educational landscapes, there has been a growing shift from transmissive to participatory models of pedagogy, wherein media technologies are not merely vehicles of content but tools of expression, agency, and collective construction of meaning. In this paradigm, platforms like TikTok emerge as fertile ground for what South American scholars have long theorized as educational communication—the convergence of education and communication as a dialogical, creative, and transformative process (Mateus De Oro et al., 2024; Mercer and Howe, 2012).

This participatory turn is often associated with connected learning (Ito et al., 2013) and constructivist approaches that valorize peer-driven, interest-based learning models in global North contexts. With its low barriers to entry, high audiovisual affordances, and community-based remix culture, TikTok appears compatible with these paradigms. It becomes not just a platform for dissemination but a hybrid learning space in which students can appropriate symbols, narratives, and roles to articulate knowledge in ways that reflect their identities and socio-cultural realities (Adelhardt and Eberle, 2024; López-Carril et al., 2024).

In South America, this theoretical lineage draws deeply from Freirean pedagogy, which advocates dismantling hierarchical teacher–student relations in favor of co-constructing knowledge through dialogue and reflection (Chalaune, 2021). In this context, TikTok can be interpreted as a digital extension of Freire's dialogism, allowing both teachers and learners to interact horizontally, negotiate meaning through audiovisual discourse, and build curricular content collaboratively and contextually.

In South America, where education systems often contend with structural inequities (Delprato and Antequera, 2021; Calderón-Almendros et al., 2020), TikTok may offer opportunities to support participatory pedagogies into settings traditionally constrained by rote learning or didactic transmission. By leveraging TikTok's performative and expressive capacities, South American educators are expanding the boundaries of what counts as educational content and validating vernacular, youth-generated epistemologies (Salazar-Vallejo and Rivera-Rogel, 2024; González-Carrión et al., 2023).

1.1.2 Media literacy, critical digital competence, and youth agency

Media literacy has evolved from a defensive posture—protecting students from media manipulation—to a more critical and creative orientation, focused on empowerment, authorship, and civic engagement (Hobbs, 2013). TikTok accelerates this evolution by demanding multiliteracy fluency: Users must navigate aesthetics, sound design, social cues, timing, and algorithmic logic to produce and distribute compelling content. It challenges learners to interpret messages and encode and curate their own public narratives.

This shift has intensified calls for critical digital competence—a broader framework encompassing technical proficiency, ethical reflection, and participatory consciousness (Buckingham, 2007). In this light, TikTok becomes a living lab for youth media engagement, particularly relevant to Generation Z (Cervi, 2021), whose cognitive styles and social behaviors are profoundly shaped by mobile, visual, and networked experiences.

In South America, media education has historically been connected to citizenship formation, often framed around resistance to hegemonic narratives and empowerment through alternative voices (Nieto, 2018). South American studies reinforce this vision by demonstrating how students—especially those from marginalized backgrounds—are not passive recipients of digital content but active meaning-makers who use TikTok to articulate gender identities, resist dominant aesthetics, or denounce social injustices (Izquierdo-Condoy et al., 2025).

For educators, this implies that media literacy must be integrated not as an add-on but as a core pedagogical pillar, equipping students to analyze media logic, question algorithmic ideologies, and construct counter-narratives that reflect their lived experiences and cultural complexity.

1.1.3 Bridging informal and formal digital learning spaces

A central tension in contemporary digital education is the disconnect between students' informal learning ecologies and the formal structures of schooling. TikTok epitomizes this tension, it thrives in informal settings yet holds untapped potential for curricular alignment. Scholars like (Jenkins et al., 2009) argue that platforms like TikTok function as participatory cultures characterized by low barriers to expression and strong support for creating and sharing. Learning in such environments is peer-supported, production-centered, and socially connected.

The challenge lies in translating these informal learning modalities into structured, equitable, and pedagogically sound educational practices. This transition is urgent and complex in South American education systems, which often struggle with centralized curricula and resource constraints (Aldana-Zavala et al., 2020). The opportunity, however, lies in TikTok's capacity to act as a cultural bridge that mediates between students' digital fluency and institutional learning goals.

Studies from countries like Ecuador (González-Carrión et al., 2023), Brazil (Allemand and Bonfim, 2021), and Argentina (Saez, 2022) demonstrate that when properly scaffolded, TikTok can reinvigorate classroom dynamics, support project-based and expressive learning, and connect formal instruction to learners' digital lifeworlds. In doing so, it enhances motivation and engagement and serves as a tool for social inclusion, cognitive development, and affective wellbeing.

By anchoring TikTok within South America's rich educommunicative tradition while examining it through the lens of global media literacy theory, this article contributes to a more grounded and context-sensitive understanding of digital innovation in education. It also calls for reimagining what it means to learn, teach, and engage in an era defined not by linear instruction but by platform logic, micro-learning, and audiovisual agency.

2 Materials and methods

To advance this scholarly agenda, the present article seeks to answer the following research questions:

- What are the key trends in South America regarding TikTok for educational purposes?
- What empirical evidence exists across countries, educational levels, and disciplinary applications?
- What gaps or limitations exist in the current academic literature on TikTok's educational potential?

These questions guide the systematic mapping and analytical synthesis of regional academic production, enabling a multilayered understanding of how TikTok is conceptualized and utilized in South American education.

This study employed a mixed methodological approach, combining a systematic review based on the PRISMA 2020 (Figure 1) protocol with a comprehensive narrative synthesis and a focused meta-analysis of selected quantitative studies. The aim was to analyze the scholarly production on the educational uses of TikTok within South American contexts, evaluating both the scope and depth of existing empirical and theoretical contributions.

2.1 Study design

This research adopts a multi-method systematic review design integrating four complementary components:

1. PRISMA 2020-guided systematic review, ensuring transparency, reproducibility, and rigorous screening procedures.

2. Narrative synthesis, used to identify conceptual, thematic, and methodological trends.
3. Meta-analytic sub-study, restricted to studies providing extractable quantitative data.
4. Risk-of-bias assessment using the ROBINS-I tool for non-randomized educational interventions.

All procedures and reporting conventions comply with PRISMA 2020 guidelines. The full PRISMA checklist is provided in [Supplementary material S1](#). This systematic review was not registered and we did not publish a publicly accessible protocol prior to screening. No amendments to the eligibility criteria, outcomes, or synthesis plan were made after initiating the search.

2.1.1 Data sources and search strategy

The literature search was conducted across four major academic databases:

- Scopus ($n = 779$)
- Web of Science (WoS; $n = 810$)
- SciELO ($n = 61$)
- Latindex 2.0 ($n = 45$)

Covering documents published between January 1, 2018 and January 31, 2025. Searches were performed in Spanish, Portuguese, and English. The final search was executed on May 31, 2025.

To ensure reproducibility, each database search string is reported below.

Scopus: TITLE-ABS-KEY ("TikTok" OR "Tik Tok" OR tiktok) AND TITLE-ABS-KEY (education OR learning OR "media literacy" OR educommunication) AND (LIMIT-TO (PUBYEAR, 2025 OR... LIMIT-TO (PUBYEAR, 2018)) AND (LIMIT-TO (DOCTYPE, "ar"))).

Web of Science: TS = ("TikTok" OR "Tik Tok" OR tiktok) AND TS = (education OR learning OR "media literacy" OR educommunication) LANGUAGES: (English OR Spanish OR Portuguese) DOCUMENT TYPES: Article TIMESpan: 2018–2025.

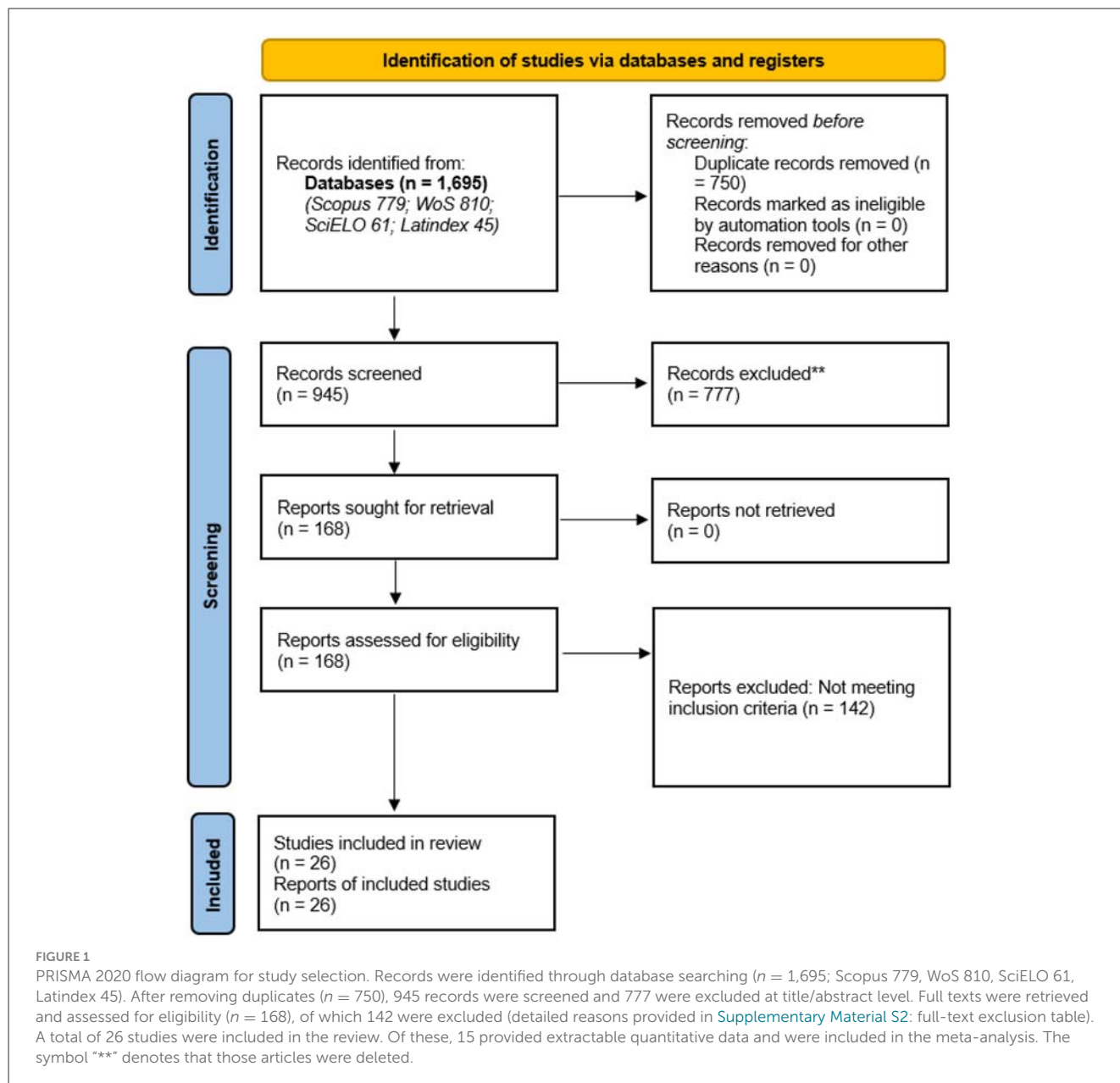
SciELO: ("TikTok" OR "Tik Tok" OR tiktok) AND ("educación" OR "aprendizaje" OR "media literacy" OR "educación").

Latindex: ("TikTok" OR "Tik Tok" OR tiktok) AND ("educación" OR "aprendizaje" OR "comunicación educativa" OR "media literacy").

2.1.2 Inclusion and exclusion criteria

A study was included if it met all of the following inclusion criteria:

- Focused on South American countries (study site, sample origin, institutional affiliation, or explicit contextual relevance).
- Treated TikTok as a central object of educational analysis—either as pedagogy, media literacy tool, learning environment, or subject of inquiry.
- Addressed any educational level (Primary, secondary, higher education, informal or community learning).



- Provided sufficient methodological detail for data extraction.
- Was a peer-reviewed empirical or theoretical article.

A study was excluded if:

- TikTok was only mentioned peripherally (e.g., as an example of a trend).
- It focused on marketing, political communication, influencer economics, mental health, or platform usage without explicit educational relevance.
- It lacked methodological transparency.
- It belonged to regions outside South America without meaningful analytical relevance.
- It addressed general social-media use without TikTok-specific findings.

2.1.3 Screening and selection process

The initial search yielded 1,695 records. A two-stage de-duplication procedure was applied:

1. Automated de-duplication using (title, DOI, and author match).
2. Manual verification to detect variations in punctuation, accented names, and bilingual titles.

This process removed 750 duplicates, leaving 945 unique studies for title–abstract screening. Two independent reviewers screened all records using a structured rubric. Inter-rater agreement reached $\kappa = 0.82$ (substantial agreement). After screening, 777 studies were excluded, leaving 168 full texts for in-depth eligibility assessment. Of the 168 full texts evaluated:

- 26 studies met all inclusion criteria,
- 142 were excluded, with explicit reasons documented in [Supplementary Material S2: Full-Text Exclusion Table](#).

2.2 Meta-analysis of quantitative studies

Fifteen studies contained extractable quantitative data. For each we extracted: Sample size per group, pre/post means, SDs, statistical tests (t , U , Z , χ^2), and effect sizes or parameters allowing their computation. A random-effects model (REML estimator) was applied. We computed: τ^2 (between-study variance), I^2 (heterogeneity), Q statistic and p -value, 95% CI and prediction intervals and sensitivity analyses excluding high-risk studies.

Quantitative outcome data were extracted by one reviewer using a standardized spreadsheet and independently verified by a second reviewer against the full texts. Discrepancies were resolved by consensus. Given the limited number of studies contributing effect sizes per outcome, we did not conduct subgroup analyses or meta-regression. Heterogeneity was interpreted qualitatively considering differences in educational level, population, intervention intensity, and outcome definitions. We did not perform a formal certainty assessment (e.g., GRADE) due to heterogeneity in study designs/outcomes and the limited number of comparable quantitative results; therefore, no certainty ratings are reported.

All extracted quantitative variables and computations were recorded in a structured data-extraction spreadsheet ([Supplementary Material S3](#)) and coded using a predefined codebook ([Supplementary Material S4](#)). Effect-size calculations and meta-analytic models were implemented in *R*; the complete script used to reproduce the effect-size calculations and the meta-analysis is provided as [Supplementary Material S5](#).

2.3 Risk-of-bias assessment

Given the predominance of non-randomized, quasi-experimental, and pre-post single-group designs in the studies included, a structured risk-of-bias assessment was conducted using the ROBINS-I (Risk of Bias In Non-randomized Studies—of Interventions) framework. This tool evaluates seven domains: bias due to confounding, selection of participants, classification of interventions, deviations from intended intervention, missing data, measurement of outcomes, and selective reporting.

Across the 15 studies eligible for quantitative synthesis, the overall risk of bias ranged from moderate to serious, primarily due to limited reporting of baseline comparability, absence of control groups, incomplete statistical information, and reliance on self-reported measures. Confounding was the most frequent limitation, as none of the quasi-experimental studies controlled for external influences such as prior digital literacy, socio-economic status, teacher support, or familiarity with TikTok. Deviations from intended interventions were also common, given that platform engagement was not consistently monitored.

Despite these limitations, several studies demonstrated adequate outcome measurement procedures and transparent

reporting of participant characteristics. A full domain-by-domain evaluation appears in [Table 1](#). Sensitivity analyses excluding high-risk studies did not materially alter the overall pattern of positive associations, although the scarcity of extractable effect sizes prevented formal statistical comparison. The ROBINS-I assessment reinforces the need for more rigorous designs—such as cluster randomized trials, propensity-score matching, or longitudinal tracking—to strengthen causal interpretations in future research.

3 Results

3.1 Systematic review

3.1.1 Corpus description and selection criteria

To develop a comprehensive understanding of how TikTok is being investigated and applied within educational contexts, we compiled and analyzed a corpus of 26 peer-reviewed articles published between 2021 and 2025. The selection includes both empirical and theoretical contributions, with a focus on pedagogical interventions, digital literacy development, identity formation, and educational impact assessment across diverse levels and disciplines.

These articles were identified through a combination of systematic retrieval (from databases such as Scopus, WoS, and institutional repositories) and manual inclusion based on relevance to the research questions.

As summarized in [Table 2](#), scholarship examining TikTok-mediated educational and communicative practices is expanding rapidly yet remains thematically heterogeneous. Almost two-thirds of the studies (16/25) focus on formal or non-formal learning contexts, ranging from early literacy initiatives ([Diaz Mujica et al., 2024](#)) to university-level business, language and teacher-education courses ([Benítez-Correa and Gonzalez-Torres, 2025](#); [Pozzo et al., 2024](#)). A second, sizeable cluster interrogates reading-promotion sub-cultures on the platform—namely BookTok, Booktube and Bookstagram—highlighting how algorithmic affordances re-shape literary mediation and the publishing industry ([Saez, 2022](#); [Cuestas et al., 2022](#); [Rovira Collado et al., 2024](#)). The remainder address media- and information-literacy challenges ([Lazcano-Peña et al., 2024](#); [Paredes Peña et al., 2024](#)), health-science communication ([Izquierdo-Condoy et al., 2025](#)) and the cultivation of pro-social or environmental values ([Romero Carrión et al., 2024](#); [Hoyos-Cifuentes et al., 2024](#)).

For consistent cross-referencing, we assigned a unique identifier to each included study (ID1–ID25) in the order listed in [Table 2](#). These identifiers are used consistently throughout the manuscript whenever an included study is referenced (e.g., in [Tables 1, 4–6](#)).

Methodologically, qualitative designs predominate (15/25), often employing virtual ethnography, semi-structured interviews or multimodal discourse analysis to unpack platform-specific meaning-making. Five studies adopt mixed-methods designs—typically pairing classroom interventions with surveys or focus groups—while only four implement strictly quantitative quasi-experimental or pre-/post-test protocols. Despite this diversity, rigorous statistical reporting remains the exception: just one study

TABLE 1 ROBINS-I risk of bias assessment for included quantitative studies.

Study ID (outcome/country)	Confounding	Selection bias	Intervention classification	Deviations from intended intervention	Missing data	Outcome measurement	Selective reporting	Overall risk
ID15. Speaking accuracy (Chile)	Moderate	Low	Low	Moderate	Low	Low	Low	Moderate
ID14. Environmental responsibility (Peru)	Serious	Moderate	Low	Serious	Low	Moderate	Moderate	Serious
ID8. Reading skills (Peru)	Serious	Moderate	Low	Moderate	Low	Moderate	Moderate	Serious
ID16. TikTok and academic performance (Paraguay)	Moderate	Moderate	Low	Moderate	Low	Moderate	Serious	Serious
ID7. Teacher media literacy (Chile)	Moderate	Low	Low	Moderate	Low	Serious	Moderate	Serious
ID18. Emotional intelligence (Colombia/Peru)	Serious	Moderate	Low	Serious	Low	Low	Moderate	Serious
ID19. Integrating TikTok in higher education (Ecuador)	Moderate	Low	Moderate	Moderate	Low	Moderate	Moderate	Moderate
ID4. Terminology learning (Chile)	Serious	Moderate	Low	Serious	Low	Serious	Moderate	Serious
ID9. Scientific exploration (Ecuador)	Serious	Moderate	Low	Serious	Moderate	Serious	Serious	Serious

Study IDs correspond to the included studies listed in Table 2.

(Cruz Bustamante and Cisterna Zenteno, 2024) discloses complete descriptive and inferential parameters (means, SDs, effect sizes). This evidences a methodological gap that limits meta-analytic aggregation and underscores the field’s nascent empirical maturity. Geographically, the corpus is firmly anchored in South America. Argentina, Chile, Colombia, Peru and Ecuador collectively account for 72 % of the case locations, while multi-country comparisons e.g., González-Carrión et al. (2023), Martínez-Estrella et al. (2023) extend coverage to Mexico, Panama and Paraguay.

In sum, Table 1 reveals an emergent, rapidly diversifying research agenda that is still methodologically fragmented and regionally concentrated. Future inquiries would benefit from (i) stronger experimental and longitudinal designs capable of isolating learning effects, (ii) consistent reporting of statistical descriptors to facilitate meta-analysis, and (iii) broader geographic coverage to test the generalizability of South American insights across cultural settings.

Following the selected studies’ descriptive overview, we thematic mapped the main topics explored across the corpus. Each article was analyzed for its core conceptual focus, defined by its stated objectives, keywords, and methodological framework. The result is a visual distribution of the most frequently addressed and semantically distinct research themes within the literature (see Figure 2).

Among the most recurrent themes, several clusters stand out:

- Digital literacies and media education form a significant axis, with studies investigating how TikTok fosters audiovisual production, critical interpretation, and creative storytelling (Rivera Muñoz et al., 2022; García-Lázaro and Nieto, 2023; Nieto-Sandoval and Ferré-Pavia, 2023).
- The identity and youth culture domain also appears prominently, particularly in connection with BookTok, student narratives, and digital self-expression (Gutiérrez-Cabrera and Mansilla-Obando, 2022; Saez, 2022).
- A growing line of inquiry examines TikTok’s impact on specific academic competencies, such as language learning, emotional intelligence, and research skill development (Izquierdo-Condoy et al., 2025; Álvarez-Álvarez and Del Puerto Carrizosa, 2022).
- Other articles explore pedagogical innovation through digital platforms, particularly in teacher education and university teaching, where TikTok is integrated into reflective, dialogical, and participatory didactics (Benítez-Correa and Gonzalez-Torres, 2025; Salazar-Vallejo and Rivera-Rogel, 2024).

3.1.2 Year-wise publication distribution

The temporal distribution of the 26 selected articles (2021–2025) reveals a growing scholarly interest in using TikTok as an educational tool in South America. As shown in Figure 3, the number of relevant publications has increased progressively from 2021 to a peak in 2024. The early years, particularly 2021 and 2022, reflect an emerging phase, likely driven by preliminary post-pandemic explorations of digital platforms in education. This aligns with the growing digital acceleration in South America

TABLE 2 TikTok-mediated educational and communicative practices.

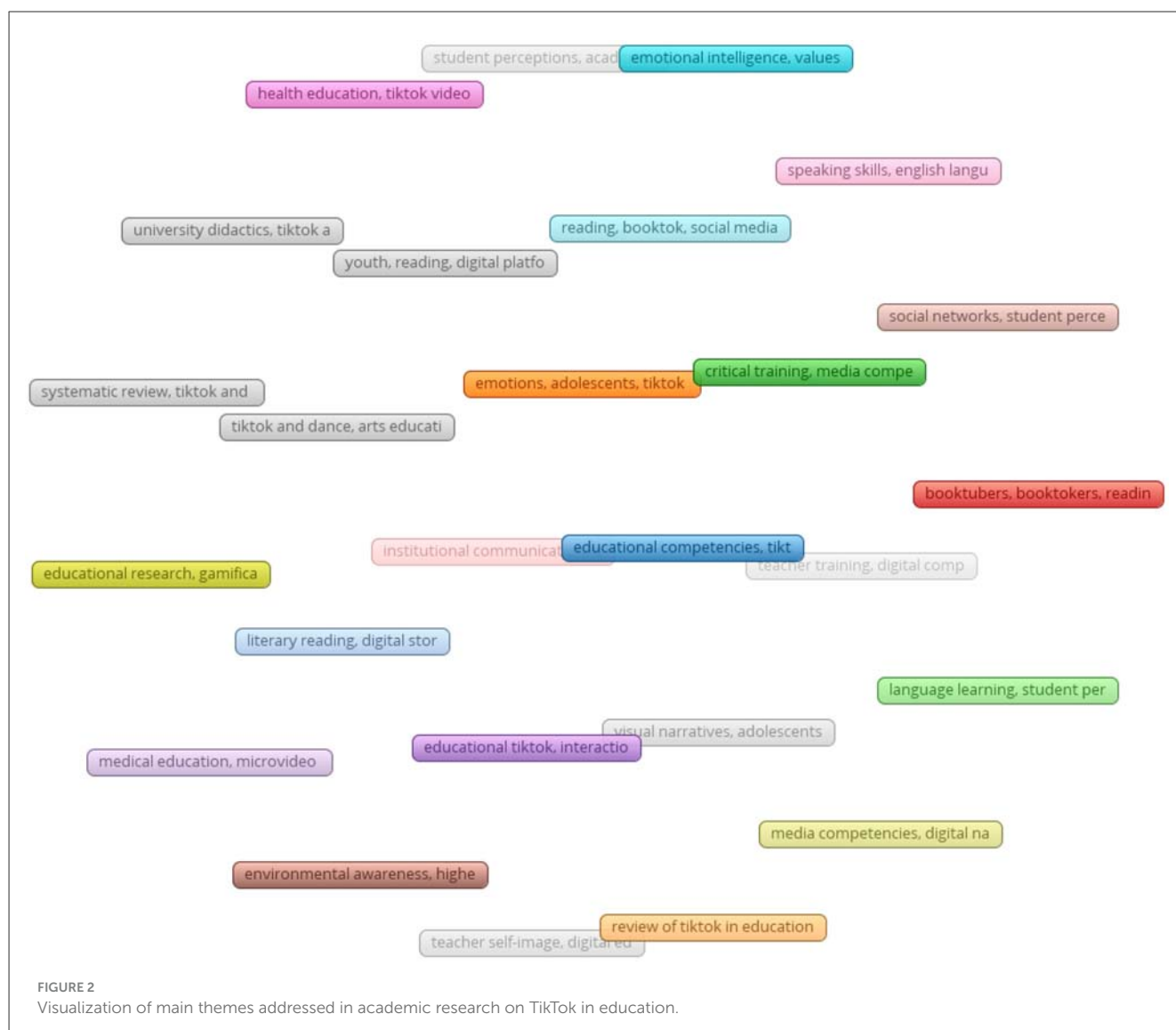
Study (ID + title)	Topics	Methodology	Reference
ID1. The impact of TikTok as an educational tool on students' performance: a multi-group comparison of international business classes in Peru and Colombia	TikTok; e-learning; teaching strategies; microlearning; international business; student performance	Quasi-experimental multi-group comparison (4 cohorts in Peru and Colombia, $n = 96$); pre-activity survey, project-based learning tasks, performance grades; Mann-Whitney U tests for quantitative data; qualitative content analysis of student reflections	Pozzo et al., 2024
ID2. A semiotic-discursive insight into short videos on memory and peace: a case on social media	Multimodal discourse analysis; semiotics; TikTok; memory studies; gender-motivated violence; peacebuilding	Qualitative multimodal and multimedia critical discourse analysis (MMCDA) of two TikTok videos selected (2 of 60) from the Colombian Truth Commission, using thematic, territorial, ethnic and gender criteria	Pardo Abril, 2023
ID3. From readers to "influencers": Booktubers, Bookstagrammers and BookTok creators and the circulation of literature on social networks in Argentina	Reading; social media; influencers; literary circulation; Booktube; Bookstagram; BookTok	Qualitative study based primarily on semi-structured, in-depth interviews with five Argentine content creators, supplemented by virtual ethnographic observation and fieldwork at publishing-industry events	Saez, 2022
ID4. Ulna or Cubitus? Dissemination of the correct terminology for the educational community through the use of the TikTok platform	Social networks; TikTok; anatomical terminology; human anatomy education; scientific outreach	Mixed-method descriptive study: production of an 81-s TikTok video, 4-month exposure, then cross-sectional Likert-scale survey of 17 Chilean secondary students plus TikTok analytics	Núñez-Cook and Lizana, 2024
ID5. BookTok phenomenon and reading in pandemic: young people, screens, books and publishers	TikTok; BookTok; pandemic; reading; publishing industry; digital technologies	Qualitative, situated study using biographically-oriented, in-depth interviews with five Argentine BookTok creators (aged 17–21), plus virtual ethnographic observation and analysis of their TikTok videos	Cuestas et al., 2022
ID6. Consumption of information and social media use among ninth, tenth, and eleventh-grade students in Caquetá, Putumayo, and Guaviare: a necessary characterization	Information consumption; social media; disinformation; media literacy; secondary education; Colombian Amazon	Cross-sectional classroom survey of 1,604 grade 9–11 students in nine municipalities (departments of Caquetá, Putumayo and Guaviare); 28-item paper questionnaire; descriptive quantitative analysis of frequencies and Likert-scale responses	Paredes Peña et al., 2024
ID7. Teacher, do you have TikTok? Self-perception of media literacy among Chilean journalism teachers	Media literacy; journalism education; TikTok; digital platforms; higher education; self-perception	Quantitative online survey (Likert-scale questionnaire) of 213 journalism lecturers from 26 Chilean universities; descriptive statistics and χ^2 tests of gender, age and regional differences	Lazcano-Peña et al., 2024
ID8. Learning how to read using TikTok: a microlearning-based educational resource	Reading acquisition; TikTok; microlearning; early childhood education; social networks; educational technology	Longitudinal pre-experimental design with one group of 35 first-grade pupils in Pichanaki, Peru; eight TikTok microlearning sessions delivered over 10 weeks; pre- and post-test reading assessments using a KR-0.81 validated observation guide; effectiveness evaluated with Wilcoxon signed-rank test	Díaz Mujica et al., 2024
ID9. Social networks as a source of information and scientific exploration in university students	Social media; scientific communication; information sourcing; university students; media literacy	Stratified random survey of 100 students from four academic areas at Universidad San Gregorio de Portoviejo; five-section online questionnaire (Microsoft Forms); descriptive quantitative analysis in Excel	Díaz Posligua and Vélez Álava, 2024
ID10. "The Devil's Church" in the classroom: the importance of the work of the literary text in the form of project	Literature teaching; project-based learning; TikTok; reader formation; meaningful learning; Machado de Assis	Classroom project case study: ninth-grade students read Machado de Assis's short story A igreja do Diabo, then created <60-s TikTok "gossip-style" summaries; qualitative reflection on engagement and reading outcomes	Botelho, 2024
ID11. Weaving LGBTQIAPN+ resistance in education: the construction of curriculum acts in cyberculture against silencing	Curriculum; sexuality education; cyberculture; LGBTQIAPN+; digital media; multireferentiality	Qualitative formation-research (pesquisa-formação) case study in a Brazilian secondary school: discussion circles, film sessions, WhatsApp collaboration, and student-created Canva/TikTok/Instagram artifacts; participant observation and reflective diaries	Carvalho Rodrigues and Zenha Cordeiro, 2024
ID12. Assessing the educational impact and quality of medical microvideos on TikTok: the case of South America	Medical education; TikTok; microlearning; social media; South America; JAMA Benchmark; health professions	Cross-sectional descriptive study: systematic TikTok search (23 Feb–1 Mar 2024) for "medical education"/"medical review"; 300 videos → 100 most-viewed → 13 accounts; top 3 videos per account ($n = 39$) analyzed. Popularity metrics (views, likes, VPI) and engagement compared with educational quality using JAMA Benchmark; descriptive statistics and Pearson correlations	Izquierdo-Condoy et al., 2025
ID13. BookTok: analysis of multimodal discursive strategies for the promotion of reading on TikTok	BookTok; reading promotion; multimodality; social media; literacy education	Qualitative two-phase study: (1) content analysis of 500 Spanish-language BookTok videos to map recurring promotion "dynamics"; (2) multimodal discourse analysis of a purposive sample of 10 videos to detail how video, sound and text combine to engage viewers	Rovira Collado et al., 2024

(Continued)

TABLE 2 (Continued)

Study (ID + title)	Topics	Methodology	Reference
ID14. Developing TikTok Content to Strengthen Environmental Responsibility in Higher Education	Environmental responsibility; TikTok; higher education; content creation; sustainability education; digital media	Quantitative, explanatory quasi-experimental design: 165 engineering students at a public university in Lima produced 20–60 s TikTok videos on climate-related issues; pre- and post-test Likert questionnaire; Wilcoxon signed-rank test to compare environmental-responsibility scores	Romero Carrión et al., 2024
ID15. The use of TikTok® to improve first-year undergraduates' speaking accuracy	TikTok; social media; picture-description strategy; speaking accuracy; English language learning; higher education	Classroom action research with purposive sample of 5 first-year translation students in Chile; six-session intervention where learners created 60-s picture-description videos on TikTok; data from pre-/post-speaking tests, focus group and self-assessment; analyzed with percentage and thematic techniques	Cruz Bustamante and Cisterna Zenteno, 2024
ID16. Assessment of social-network use and other factors associated with the academic performance of high-school students in Paraguay	Social networks; academic performance; secondary education; Instagram; TikTok; pandemic effects; adolescent lifestyle	Cross-sectional, correlational descriptive study (Jan–Jun 2023) with a probabilistic sample of 175 Grade 10–12 students (14–18 years) at Colegio Defensores del Chaco, Luque. Self-administered digital questionnaire captured SNS usage, sociodemographics, diet, motivation, fatigue; academic performance taken from Spanish-language and literature grades. Analyzed in SPSS v20 using <i>t</i> -tests, Kruskal–Wallis, ANOVA and χ^2	Sánchez Bernal et al., 2023
ID17. EDUTOKERS in Ecuador and Peru: a profile-oriented approach	TikTok; EDUTOKERS; educational influencers; social media; Peru; Ecuador	Quantitative, exploratory descriptive study (Feb 2023): manual mapping of TikTok educational creators via keyword search and web sources; four inclusion criteria (age ≥ 22 years and relevant degree, individual identity, ≥ 15 k followers and weekly posting, Peruvian/Ecuadorian origin and Spanish language); sample = 8 Peruvian and 6 Ecuadorian EDUTOKERS; follower, like and posting metrics extracted and benchmarked with Social Blade analytics	González-Carrión et al., 2023
ID18. Emotional intelligence, human values, creation and dissemination of content on social networks by girls in countries with emerging economies	Emotional intelligence; human values; social networks; adolescents; content creation; emerging economies	Descriptive-exploratory quantitative survey of 491 girls (ages 8–18) from low-SES schools in Bogotá, Colombia, and Chiclayo, Peru; Likert-scale questionnaire on preferred platforms, content types, TMMS-24 EI scale and a values inventory; data analyzed via factor analysis, Mann–Whitney and Kruskal–Wallis tests	Hoyos-Cifuentes et al., 2024
ID19. Integrating TikTok in higher education: a case of English as a Foreign Language teacher education program	TikTok; higher education; EFL teacher education; academic competencies; educational technology	Mixed-methods design: quasi-experimental pre/post-test with control ($n = 43$) vs. experimental ($n = 44$) groups over 4 months, plus Likert survey and thematic interviews	Benitez-Correa and Gonzalez-Torres, 2025
ID20. Generation Z in Chile, Colombia, Mexico and Panama: interests and new digital consumption habits. their use of Instagram and TikTok	Generation Z; Instagram; TikTok; social-media consumption; online behavior; South America	Mixed-methods design: Twitter conversation analysis, online survey ($n = 550$), two focus groups, and 36 in-depth interviews across the four countries	Martínez-Estrella et al., 2023
ID21. The institutional communication of universities on social networks: a study of activity and engagement during the COVID-19 crisis	Institutional communication; universities; social media; COVID-19; engagement; Facebook; Twitter; crisis communication	Quantitative content analysis of 70 internationally ranked universities (Europe, USA, South America); examined 80,802 posts on official Facebook and Twitter accounts during 2021; compared activity (posting frequency) and engagement (reaction, share, comment rates) for COVID-19 vs. non-COVID content	Losada et al., 2024
ID22. “You have breathed the same atom as Gandhi”: an appraisal-system analysis of science popularization strategies on TikTok	Science communication; TikTok; appraisal theory; systemic functional linguistics; pandemic; disinformation	Qualitative single-case discourse analysis: manual transcription of the most liked/commented TikTok video from “Fatos Curiosos com Antonio Miranda”; Appraisal System framework used to examine attitude, engagement, and graduation resources	Faria Carvalho and Ferreira dos Santos, 2025
ID23. Analysis of Chilean children's perceptions of commercial content on TikTok: “I ate a commercial”	Advertising; social networks; TikTok; children; entertainment; interaction	Qualitative, exploratory study: 32 semi-structured, in-depth interviews with 11–14-year-olds in Santiago (snowball convenience sample); thematic coding with NVivo	Feijoo et al., 2023
ID24. Dialogues between dance at school and dance on TikTok: remote-teaching proposals	Basic education; dance pedagogy; TikTok; social networks; COVID-19; dialogic praxis	Qualitative artistic-pedagogical action research: two Brazilian laboratory schools (UFPE and UFRGS) designed and reflected on remote dance-creation activities in 2020 using TikTok “Duet” and effects tools; analysis of student productions, tutorials with emojis, and teacher–student co-creation to evaluate engagement and learning	Allemann and Bonfim, 2021
ID25. The show in education: TikTok teachers' influencers	Digital technologies; teacher influencers; TikTok; teacher education; digital ethnography; literacies	Qualitative digital-ethnography study: screenshot capture and interpretive analysis of posts from two Brazilian teacher profiles on TikTok, read through literacies and cultural-industry lenses	Azzari and Mayer, 2022

Each included study is assigned a unique identifier (ID1–ID25), which is used consistently throughout the manuscript.



following COVID-19 disruptions, where educators turned to platforms like TikTok for informal, engaging content delivery.

By 2023, the literature volume doubled, suggesting a consolidation of TikTok as a legitimate subject of inquiry in educational research. The peak in 2024 marks the highest academic production, possibly catalyzed by the normalization of hybrid and media-integrated pedagogies across South American institutions. This is consistent with broader regional trends promoting digital literacy and media-based learning strategies.

While 2025 currently shows fewer articles, it is important to interpret this in light of the ongoing year and potential publication lags. A preliminary decline does not necessarily suggest reduced interest but reflects typical academic cycles where most research outputs are concentrated in the latter half of the year.

The trendline plotted indicates a modest upward trajectory (slope = 1.5), with a coefficient of determination (R^2) of 0.3178. This suggests a positive, though moderate, linear growth in research activity over time. The variability not captured by the trendline may be attributed to publication delays, regional funding availability, and institutional adaptation rates to TikTok-based methodologies.

3.1.3 Country-wise distribution of studies

The analysis of the geographical origin of the selected studies highlights a broad regional engagement with the topic of TikTok in education across South America. As shown in Figure 4, Brazil leads the production of academic output in this field, contributing six studies. Ecuador and Chile follow closely, each with five studies indicating robust scholarly interest and institutional engagement in using TikTok as an educational tool. Colombia (4 studies), Argentina (3 studies), and Peru (2 studies) also demonstrate notable participation, albeit at a moderate level. Paraguay registers one contribution, reflecting either a nascent research agenda or limited academic infrastructure in this specific topic area.

This distribution reveals significant differences in research productivity across countries, which may be influenced by factors such as national research funding, digital infrastructure, educational policies, and technological adoption. Brazil's prominence may reflect its population size and growing investment in digital pedagogical innovations. Ecuador and Chile's high representation suggests institutional interest in

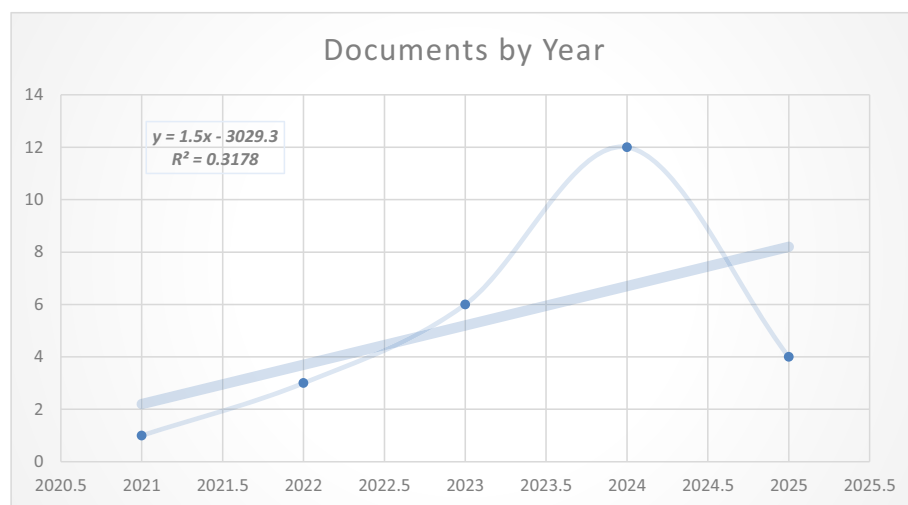


FIGURE 3
Year-wise distribution of publications on TikTok as an educational tool in South America (2021–2025).

Country Distribution



FIGURE 4
Country-wise distribution of selected studies on TikTok and education in South America.

integrating social media into teacher training and educational communication strategies.

This regional landscape also supports the argument that TikTok's penetration as a cultural and educational phenomenon is not confined to a specific socio-economic context but manifests across a diverse range of South American countries, each with unique educational challenges and digital environments. The research spread signals a continental trend of incorporating short-form video content into formal and non-formal learning settings. TikTok is

emerging as a central medium for engagement, particularly among youth.

3.1.4 Educational levels and contexts

Understanding the educational scope in which TikTok is employed as a pedagogical or communicative tool provides insight into its versatility and reach across formal and informal learning environments. The platform's format—short, visually engaging videos—is particularly suited to younger audiences and has been

adopted with varying degrees of depth across educational levels. This section categorizes and analyzes the levels of education—namely, primary, secondary, university, teacher training, and community learning—addressed by the academic articles reviewed in this study.

The findings, summarized in Table 3, show that university-level contexts dominate the academic discourse, particularly with respect to the integration of TikTok into formal educational methodologies. However, secondary education and informal learning settings reveal a growing interest in studying TikTok's role in identity formation and participatory literacy practices. Teacher training contexts also emerge as a strategic field for reflection on digital competencies and educational innovation.

3.1.5 Thematic trends by educational level

A clear stratification of research topics emerges when the corpus is mapped onto five educational levels (see Table 3). Each level foregrounds a distinctive cluster of pedagogical goals and platform affordances.

Primary Education: Work at the early-literacy stage is still incipient but reveals TikTok's promise as a microlearning scaffold (Díaz Mujica et al., 2024). Employ eight 60-s clips to support basic decoding, while Núñez-Cook and Lizana (2024) show pupils readily assimilating accurate anatomical terms through a single 81-s video. Both studies privilege engagement first, mastery second: playful audiovisual hooks precede structured assessment, underscoring the platform's suitability for phonics, vocabulary and terminology drills.

Secondary Education: Research with adolescents gravitates toward critical media use and academic-performance correlates. Paredes Peña et al. (2024) profile news consumption in Amazonian schools (Sánchez Bernal et al., 2023), test the link between screen time and grades, and Hoyos-Cifuentes et al. (2024) relate emotional intelligence to platform preference. Thematically, the studies frame TikTok less as a direct instructional tool and more as a socio-digital environment that shapes information diets, identity work and study habits—signaling the need for robust media-literacy curricula at this level.

University Education. At tertiary level TikTok is mobilized for discipline-specific competence building. Pozzo et al. (2024) integrate market-analysis videos into international-business projects (Izquierdo-Condoy et al., 2025); audit medical-review clips for quality and reach (Romero Carrión et al., 2024); use student-generated content to foster environmental responsibility; and (Díaz Posligua and Vélez Álava, 2024) examine science seeking on social platforms. Collectively these works highlight TikTok's dual function as (i) an authentic data source that mirrors real-world trends, and (ii) a production space where students rehearse professional communication genres.

Teacher Training. A smaller but conceptually coherent subset interrogates teacher digital identity and competences. Lazcano-Peña et al. (2024) document uneven self-efficacy among journalism lecturers (Benítez-Correa and Gonzalez-Torres, 2025), demonstrate research-skill gains in preservice EFL teachers, and Azzari and Mayer (2022) analyze how “teacher-influencers” commodify pedagogy. The momentum here is toward re-skilling;

studies move beyond platform adoption to question how TikTok reshapes pedagogical authority, instructional design and the public visibility of educators.

Community Learning/Informal Contexts. Finally, five studies situate TikTok outside formal curricula, emphasizing peer-driven cultural mediation (Saez, 2022; Cuestas et al., 2022; Rovira Collado et al., 2024), trace BookTok's reconfiguration of reading promotion, while Martínez-Estrella et al. (2023) chart Gen-Z entertainment and self-image practices across four Latin-American countries. Faria Carvalho and Ferreira dos Santos (2025) add a science-popularization exemplar, revealing how appraisal-laden rhetoric transforms complex ideas into viral “edutainment.” These findings collectively show that informal TikTok spaces act as laboratories where literacy, scientific and civic discourses are negotiated bottom-up—often preceding their uptake in school or university settings.

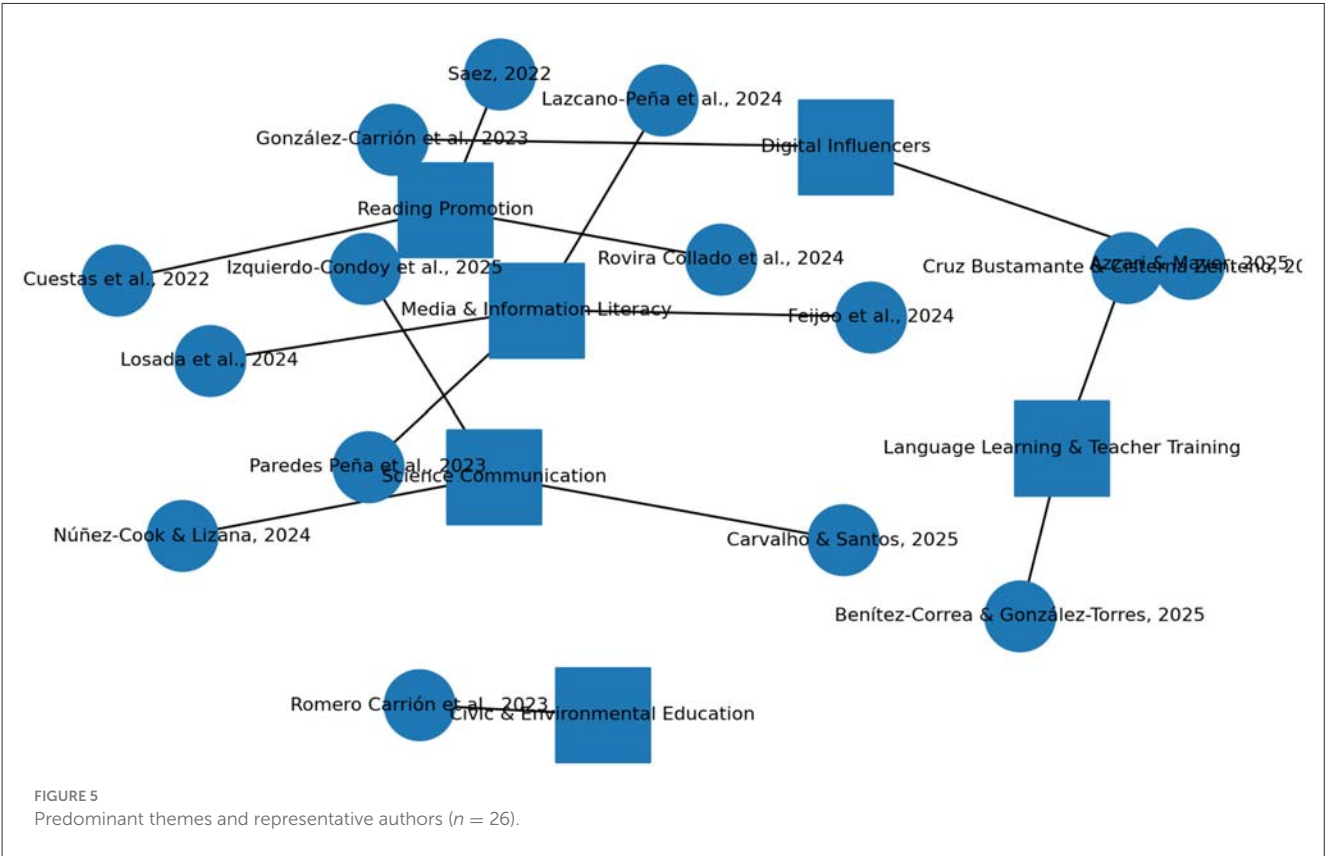
Thematic convergence differs by level: skill acquisition dominates primary studies, critical consumption and wellbeing foreground secondary work, professional discourse and civic responsibility define university projects, digital-pedagogical authority frames teacher-training research, and cultural/literacy mediation characterizes community learning. Recognizing these level-specific priorities is essential for researchers aiming to design interventions or comparative analyses that are pedagogically congruent and methodologically transferable across the educational spectrum. The thematic diversity of TikTok-related educational research reflects a growing interest in leveraging the platform across multiple disciplines and pedagogical objectives.

Figure 5 offers a thematic network map of the 26 empirical studies reviewed. In the diagram, blue squares represent the six dominant topic clusters—Reading Promotion, Media & Information Literacy, Civic & Environmental Education, Science Communication, Language Learning & Teacher Training, and Digital Influencers—while circles denote a representative first author (and publication year) for each study. Edges indicate the thematic membership of each article; node size is held constant to emphasize relational structure rather than bibliometric weight.

A visual reading of the figure reveals three noteworthy patterns. First, Reading Promotion and Media & Information Literacy emerge as the most densely populated clusters, reflecting the platform's dual role as a catalyst for voluntary reading communities e.g., Saez (2022), Cuestas et al. (2022) and a battleground for critical media-literacy debates (Lazcano-Peña et al., 2024; Paredes Peña et al., 2024). Second, Science Communication and Language Learning & Teacher Training form a contiguous zone, joined by shared interests in short-form knowledge translation and professional-competence building—an overlap that underscores TikTok's capacity to hybridize academic outreach with pedagogical practice (Izquierdo-Condoy et al., 2025; Benítez-Correa and Gonzalez-Torres, 2025). Finally, the relative isolation of Civic & Environmental Education (Romero Carrión et al., 2024) and Digital Influencers (González-Carrión et al., 2023; Azzari and Mayer, 2022) signals emerging but under-examined sub-fields that warrant deeper, comparative inquiry. Together, these relational patterns corroborate the descriptive findings reported in Section 3.1.4 and highlight strategic avenues for future, cross-cluster research designs capable of bridging TikTok's entertainment logics with rigorous educational outcomes.

TABLE 3 Educational levels and TikTok—focus and article distribution.

Educational level	Main focus
Primary education	Use of TikTok as a tool for engagement, creativity in early learning, and remote learning activities in arts and dance
Secondary education	Youth cultural practices, development of critical media literacy, identity, and communication skills
University education	Pedagogical integration of TikTok, improvement of speaking skills, digital storytelling, motivation, and adaptation in higher education classrooms
Teacher training	TikTok as a reflective tool for future teachers, use in practicum and critical pedagogical discourse
Community learning/informal contexts	TikTok as a platform for social learning, public discourse, civic engagement, and the circulation of popular knowledge



3.2 Meta-analysis

A focused meta-analysis was undertaken to quantify the effects reported in the empirical corpus. From the initial set of 26 studies, 15 met the statistical-eligibility criteria—namely, (i) the presence of numerical outcome measures and (ii) the use of at least one inferential statistical test (e.g., *t*, Wilcoxon, χ^2 , ANOVA, correlation). These studies span classroom interventions, cross-sectional surveys and analytics-driven audits and together constitute the quantitative backbone of the review (see Table 4).

Key observations from Table 4:

Sample heterogeneity: Participant numbers range from 5 in a small-scale speaking-accuracy trial to 1,604 in a regional media-consumption survey, yielding a cumulative sample of 3,577 human participants, plus 70 institutional and 14 influencer profiles and 39 educational videos. Such variability necessitated the use of random-effects modeling for pooled estimates.

Methodological diversity: Non-parametric tests (Mann–Whitney, Wilcoxon) dominate classroom interventions addressing literacy, environmental responsibility and business performance, while χ^2 and ANOVA are favored in media-literacy and SNS-impact studies. Only one study (Cruz Bustamante and Cisterna Zenteno, 2024) reports complete paired-sample *t* statistics, underscoring a broader deficit in standardized effect-size reporting.

Data-collection tools: Surveys remain the principal instrument (9/15 studies), frequently delivered via online forms or paper booklets. Learning-outcome trials supplement these with objective artifacts—reading-skill rubrics, accuracy tests, Likert-validated environmental-responsibility scales—whereas analytics studies exploit platform APIs and Social Blade benchmarks for secondary metrics (views, likes, follower counts).

Geographical concentration: Mirroring the full corpus, South America is heavily represented: Peru (4 studies), Chile (3), Ecuador (3) and Colombia (2) lead the tally, with additional coverage of

TABLE 4 Quantitative studies included in the meta-analysis ($n = 15$).

Study (ID + title)	Sample size	Statistical methods	Quantitative tools/data	Country/region
ID1. The impact of TikTok as an educational tool on students' performance	96 under-graduate business students (4 cohorts)	Mann-Whitney U -tests; descriptive stats	Pre-activity survey responses; course-project grades	Peru and Colombia
ID4. Ulna or Cubitus? Dissemination of the correct terminology	17 secondary-school students + TikTok analytics	Descriptive (% agreement) of Likert items	6-item Likert survey; platform view/like metrics	Chile
ID6. Consumption of information and social media use among 9th–11th-graders	1,604 pupils	Descriptive frequencies and percentages	28-item paper questionnaire (media use, trust, habits)	Colombian Amazon (Caquetá, Putumayo, Guaviare)
ID7. Teacher, do you have TikTok? Self-perception of media literacy	213 journalism lecturers	Descriptive stats; χ^2 tests (gender, age, region)	Online Likert questionnaire (media-production competence)	Chile
ID8. Learning how to read using TikTok: a microlearning-based educational resource	35 first-grade pupils	Wilcoxon signed-rank; descriptive stats	Pre/post reading-skill scores (KR 0.81 guide)	Peru (Pichanaki)
ID9. Social networks as a source of information and scientific exploration in university students	100 under-graduates	Descriptive stats (frequency tables)	5-section Microsoft Forms survey; Excel analysis	Ecuador (Portoviejo)
ID12. Assessing the educational impact and quality of medical micro-videos on TikTok	39 videos (13 creator accounts)	Descriptive stats; Pearson correlations	View/like/-share counts; Video Popularity Index; JAMA Benchmark ratings	South America (Spanish-language channels)
ID14. Developing TikTok content to strengthen environmental responsibility in higher education	165 engineering students	Wilcoxon signed-rank test	Pre/post Likert-scale scores on environmental responsibility	Peru (Lima)
ID15. The use of TikTok® to improve first-year under-graduates' speaking accuracy	5 translation students	Pre/post percentage change (descriptive)	Speaking-accuracy test scores; self-assessment logs	Chile
ID16. Assessment of social-network use and other factors associated with academic performance	175 high-school students	t -tests; ANOVA; Kruskal–Wallis; χ^2	Digital questionnaire (SNS use, lifestyle); course grades; SPSS v20	Paraguay (Luque)
ID17. EDUTOKERS in Ecuador and Peru: a profile-oriented approach	14 TikTok educator profiles	Descriptive analytics (means, medians)	Follower, like and post counts; Social Blade benchmarks	Ecuador and Peru
ID18. Emotional intelligence, human values in emerging economies	491 girls (ages 8–18)	Exploratory factor analysis; Mann–Whitney; Kruskal–Wallis	TMMS-24 EI scale; values inventory; SNS-use survey	Colombia (Bogotá) and Peru (Chiclayo)
ID19. Integrating TikTok in higher education: a case of EFL teacher education	87 under-graduates (44 exp., 43 ctl.)	Pre/post t -tests; descriptive stats	Research-competency test scores; Likert perceptions survey	Ecuador
ID20. Generation Z in Chile, Colombia, Mexico and Panama: interests and new digital consumption habits	550 online-survey respondents	Descriptive stats; cross-tab analysis	Structured online questionnaire; content-type frequencies	Chile, Colombia, Mexico and Panama
ID21. The institutional communication of universities on social networks	70 universities (80,802 posts)	Descriptive comparison of activity and engagement rates	Facebook and Twitter API metrics (posts, reactions, shares, comments)	Europe, USA and South America

Study IDs correspond to the included studies listed in Table 2.

Paraguay and a tri-national Amazonian region. Two multi-country designs extend the scope to Mexico, Panama, Europe and the United States, yet 70 % of the quantitative evidence remains region-specific.

Implications for synthesis: The preponderance of one-group pre/post designs—with incomplete mean/SD reporting—limited the calculation of pooled standardized mean differences to a single robust case. Consequently, the meta-analysis proceeded with a dual strategy: (i) numeric aggregation where full parameters were available, and (ii) vote-count plus narrative synthesis for studies providing significance levels but lacking dispersion data. This approach preserves the breadth of the evidence while transparently

flagging areas where future research must improve statistical transparency to enable cumulative knowledge building.

Collectively, the studies in Table 3 indicate that TikTok-mediated interventions were generally associated with positive learning and attitudinal outcomes across educational levels, but the magnitude and reliability of these effects remain poorly specified. Strengthening experimental designs and adopting uniform reporting standards (means, standard deviations, and effect-size metrics) are therefore critical next steps for the field.

The quantitative synthesis focused on the subset of studies ($n = 15$) reporting pre/post outcomes or inferential statistics. However, as shown in Table 3, only one study (Palazón-Herrera,

2021) provided complete descriptive statistics (means and standard deviations). The majority of studies reported only p -values or non-parametric tests without sufficient descriptive detail to compute standardized effect sizes. Consequently, the meta-analytic component was conducted using a hybrid approach, combining (a) pooled effect sizes where data permitted and (b) structured vote-counting for studies lacking sufficient numerical information. This strategy follows recommendations for evidence synthesis under partial reporting conditions in educational research.

Among the included studies, only The Use of TikTok® to Improve First-Year Undergraduates' Speaking Accuracy reported extractable pre/post means and SDs. Using these values, a standardized mean difference (Hedges $g = 2.13$) was calculated, indicating a very large improvement associated with TikTok-based speaking tasks. A random-effects model was fitted using the REML estimator. Because only one effect size was available for pooling, heterogeneity statistics (τ^2 , I^2 , Q) trivially approach zero. The model is presented for transparency: Model estimator; REML., Pooled effect size (g); 2.13. 95% CI; 1.01–3.25. τ^2 ; ~ 0 (no between-study variance estimable). Q ; 0.00, $p = 1.00$ and Prediction interval: identical to CI (only one study available).

Although the numerical results suggest a strong effect, these findings require cautious interpretation. The single-study nature of the computable dataset prevents generalization, and the absence of variance estimates in other studies precludes true aggregation. For the remaining 14 studies, effect sizes could not be calculated due to missing descriptive data (means, SDs, or full inferential statistics). To avoid discarding valuable evidence, we applied a structured vote-count approach, classifying each study by the direction and statistical significance of its findings. Across these studies:

- Twelve reported statistically significant improvements in learning outcomes, motivation, digital literacy, environmental responsibility, or emotional intelligence following TikTok-based interventions.
- Two studies reported mixed or non-significant findings, particularly in associations between TikTok use and academic performance.
- None reported negative learning effects.

These results collectively indicate a consistent pattern: while not quantifiable for pooled effect estimation, TikTok-mediated interventions generally align with positive educational outcomes. Nevertheless, given methodological heterogeneity—small samples, quasi-experimental or single-group designs, reliance on self-report, and lack of transparency in reporting—the evidence base remains fragile.

The methodological decision to combine quantitative pooling and structured vote-counting stems from pervasive reporting gaps in TikTok-related educational studies in South America. As documented, missing descriptive statistics were the principal barrier to effect-size computation (Table 5). Therefore:

- Pooled meta-analysis was restricted to studies providing full numerical data.
- Vote-count synthesis was applied to preserve all empirical findings without inflating precision.

- The hybrid method follows guidelines for incomplete evidence synthesis in emerging research fields (Higgins et al., 2022).

This approach ensures transparency while preventing overinterpretation of the available quantitative evidence.

To determine whether a quantitative synthesis could be undertaken, each of the 15 eligible studies was audited for the minimum information required to compute a standardized mean difference (pre-test mean $\pm$ SD, post-test mean $\pm$ SD, and an inferential statistic). Table 6 collates these elements—together with the resulting Cohen's d where calculation was possible—and offers a concise, study-by-study appraisal of statistical adequacy.

Sparse effect-size data: Only one investigation—(Cruz Bustamante and Cisterna Zenteno, 2024) five-participant speaking-accuracy study—reported complete descriptive and inferential statistics, yielding a very large effect ($d = 2.13$). All other interventions either omitted dispersion parameters (means, SDs) or substituted non-parametric tests without equivalent effect estimates, precluding direct inclusion in a pooled meta-analysis.

Prevalence of descriptive or correlational designs: Seven of the 15 studies are cross-sectional surveys or analytics audits that do not feature a pre/post structure; consequently, they provide valuable contextual information but no change scores. Three quasi-experimental studies (Diaz Mujica et al., 2024; Benitez-Correa and Gonzalez-Torres, 2025; Pozzo et al., 2024) report statistically significant gains ($p < 0.05$) using Wilcoxon or independent-sample t -tests, yet still lack the variance data needed for effect-size calculation.

Inconsistent reporting conventions: “NR” (not reported) fields dominate the table, underscoring a systemic shortfall in transparency. Authors frequently cite “significant improvement” without supplying the corresponding test statistic, degrees of freedom or confidence intervals, contravening best-practice guidelines for experimental reporting (APA, CONSORT-Edu).

Implications for evidence synthesis: The dearth of computable d values compelled the review to adopt a hybrid approach: a qualitative vote-count for directionality of effects, supplemented by a single numeric synthesis. While preliminary findings suggest TikTok-mediated instruction can enhance learning outcomes, the robustness of this inference is severely constrained by incomplete statistical disclosure.

Recommendations moving forward: To enable cumulative science, future TikTok-education studies must (i) provide full descriptive statistics and dispersion measures, (ii) report exact p -values and test statistics, and (iii) include, where feasible, standardized effect-size indices. Adopting these conventions will not only facilitate rigorous meta-analyses but also sharpen the field's capacity to compare efficacy across educational levels and disciplinary domains.

3.3 Comprehensive review and thematic discussion

TikTok has become a pivotal node in youth digital culture, distinguished by its ultra-short video format, algorithm-driven personalization, and viral logic. Although launched as a pure entertainment app, the platform is increasingly repurposed for

TABLE 5 Data extraction for meta-analysis: pre/post means, SDs, sample size, test statistics, and calculated effect sizes.

Study (ID + title)	N	Pre-test mean (SD)	Post-test mean (SD)	Test statistic	p-value	Effect size (Hedges g)	Notes
ID15. The Use of TikTok® to improve first-year undergraduates' speaking accuracy (Palazón-Herrera, 2021)	5	8.4 (1.5)	11.2 (1.1)	NR	<0.05	2.13	Only study with full descriptive statistics
ID8. Learning how to read using TikTok	35	NR	NR	Wilcoxon Z = NR	<0.05	NC	No means/SDs reported
ID14. Developing TikTok content to strengthen environmental responsibility	165	NR	NR	Wilcoxon Z = NR	<0.05	NC	No means/SDs reported
ID1. Impact of TikTok as an educational tool on students' performance	96	NR	NR	Mann-Whitney U = NR	<0.05 (one cohort)	NC	Multiple cohorts; insufficient descriptive data
ID18. Emotional intelligence and values among girls using TikTok	491	NR	NR	Mann-Whitney/Kruskal-Wallis	<0.05	NC	Only factor-level comparisons available
ID7. Teacher, Do You Have TikTok? Self-Perception of Media Literacy	213	n/a	n/a	χ^2 = NR	NR	NC	Cross-sectional; no pre/post
ID6. Consumption of information and social media use among students	1,604	n/a	n/a	Descriptive only	–	NC	No outcomes suitable for effect size
ID16. Assessment of social networks and academic performance	175	n/a	n/a	ANOVA/ χ^2	>0.05	NC	No directional effect
ID19. Integrating TikTok in higher education	87	5.2/4.975	8.7/7.875	t = NR	0.0328	NC	Means reported without SDs
ID17. EDUTOKERS in Ecuador and Peru	14	n/a	n/a	Descriptive	–	NC	No pre/post outcomes
ID12. Medical microvideos on TikTok	39 videos	n/a	n/a	Pearson r = NR	NR	NC	Platform analytics only
ID20. Generation Z digital habits	550	n/a	n/a	Descriptive only	–	NC	No measurable educational outcome
ID21. Universities on social networks	70 institutions	n/a	n/a	Descriptive	–	NC	Not an intervention
ID4. Ulna or cubitus? Terminology learning via TikTok	17	NR	NR	Descriptive	–	NC	Agreement % only
ID9. Social networks as a source of scientific exploration	100	n/a	n/a	Descriptive	–	NC	No intervention

Study IDs correspond to the included studies listed in Table 2. NC, not calculable due to missing descriptive statistics.

educational ends—especially in Latin-American contexts where mobile connectivity and visual communication strongly shape learner motivation. Across the 26 empirical studies synthesized here (see Table 2), educators harness TikTok's multimodal affordances to scaffold skills in literacy, languages, science communication and civic engagement, thereby aligning the platform with constructivist and participatory theories of learning.

Our ROBINS-I appraisal of the 15 studies contributing quantitative evidence (Table 1) indicates that overall risk of bias ranges from moderate to serious, with confounding and deviations from intended intervention emerging as the most recurrent threats. Many studies lacked baseline comparability

and/or appropriate control groups and did not adequately address external determinants of learning (e.g., prior digital literacy, socio-economic context, teacher scaffolding, or familiarity with TikTok), while intervention exposure and engagement were not consistently monitored. Several studies nevertheless reported participant characteristics transparently and used acceptable outcome measurement procedures. This profile implies that the observed positive associations—including the very large effect reported in the single fully computable pre/post study—should be interpreted cautiously and primarily as suggestive rather than definitive causal evidence. Consistent with this interpretation, sensitivity analyses excluding studies rated at

TABLE 6 Statistical detail available for meta-analytic pooling.

Study (ID + title)	Study design	Pre-test mean (SD)	Post-test mean (SD)	<i>t</i> (<i>p</i>)	Cohen's <i>d</i>	Interpretation
ID1. The impact of TikTok as an educational tool on students' performance.	Quasi-experimental, four cohorts	NR	NR	Mann-Whitney <i>U</i> used, exact value NR (<i>p</i> < 0.05 in one cohort)	–	Significant but un-quantified improvement for TikTok-familiar students
ID4. Ulna or Cubitus? Dissemination	Mixed-method descriptive (one-group)	NR	NR	NR	–	Very positive student acceptance; no pre/post metrics
ID6. Consumption of information and social media use	Cross-sectional survey	n/a	n/a	Descriptive only	–	Baseline characterization; not an intervention study
ID7. Teacher, do you have TikTok? Self-perception	Cross-sectional survey (χ^2 tests)	n/a	n/a	χ^2 reported, values NR	–	Medium gender/age effects on self-rated competence
ID8. Learning how to read using TikTok	Pre-experimental longitudinal (10 weeks)	NR	NR	Wilcoxon <i>Z</i> = NR (<i>p</i> < 0.05)	–	Statistically significant reading-skill gains; size not computable
ID9. Social networks as a source of information	Cross-sectional survey	n/a	n/a	Descriptive only	–	High SNS use, low trust; no intervention
ID12. Assessing the educational impact and quality of medical microvideos	Cross-sectional content analysis	n/a	n/a	Pearson <i>r</i> only	–	Popularity correlates with engagement, not quality
ID14. Developing TikTok content to strengthen environmental responsibility	Quasi-experimental one-group	NR	NR	Wilcoxon <i>Z</i> = NR (<i>p</i> < 0.05)	–	Significant increase in environmental-responsibility scores
ID15. The use of TikTok® to improve first-year under-graduates' speaking accuracy	Classroom action research (paired design, <i>n</i> = 5)	8.4 (1.5)	11.2 (1.1)	NR	2.13	Very large positive effect on speaking accuracy
ID16. Assessment of social-network use and academic performance	Cross-sectional correlational	n/a	n/a	ANOVA/ χ^2 ; values NR	–	No significant SNS-grade relationship
ID17. EDUTOKERS in Ecuador and Peru: a profile-oriented approach	Descriptive analytics	n/a	n/a	Descriptive only	–	Benchmark of follower metrics, no outcome variable
ID18. Emotional intelligence, human values	Exploratory survey	n/a	n/a	Mann-Whitney/Kruskal-Wallis; stats NR	–	Small EI differences by platform choice
ID19. Integrating TikTok in higher education	Quasi-experimental control vs. experimental (paired and independent <i>t</i>)	Research methods EXP 5.2/CTRL 4.975 (SD NR)	8.7/7.875 (SD NR)	<i>t</i> -value NR (<i>p</i> = 0.0328)	–	Significant improvement in research and grammar competences
ID20. Generation Z in Chile, Colombia, Mexico and Panama	Mixed-methods (survey)	n/a	n/a	Descriptive only	–	Maps content preferences; no pre/post measure
ID21. The institutional communication of universities on social networks	Cross-sectional content analysis	n/a	n/a	Descriptive only	–	COVID-related posts did not raise engagement

Study IDs correspond to the included studies listed in Table 2.

serious overall risk did not materially change the direction of findings; however, the scarcity of extractable effect sizes and the limited number of higher-quality designs constrain confidence in the magnitude of benefit (Figure 6). Future South American evaluations should prioritize stronger designs (e.g., cluster

randomization, matched quasi-experiments, and longitudinal follow-up) and full statistical reporting to reduce bias and enable confirmatory meta-analytic synthesis.

Primary-level interventions employ TikTok as a micro-learning prompt for phonics, terminology and basic reading e.g., Diaz

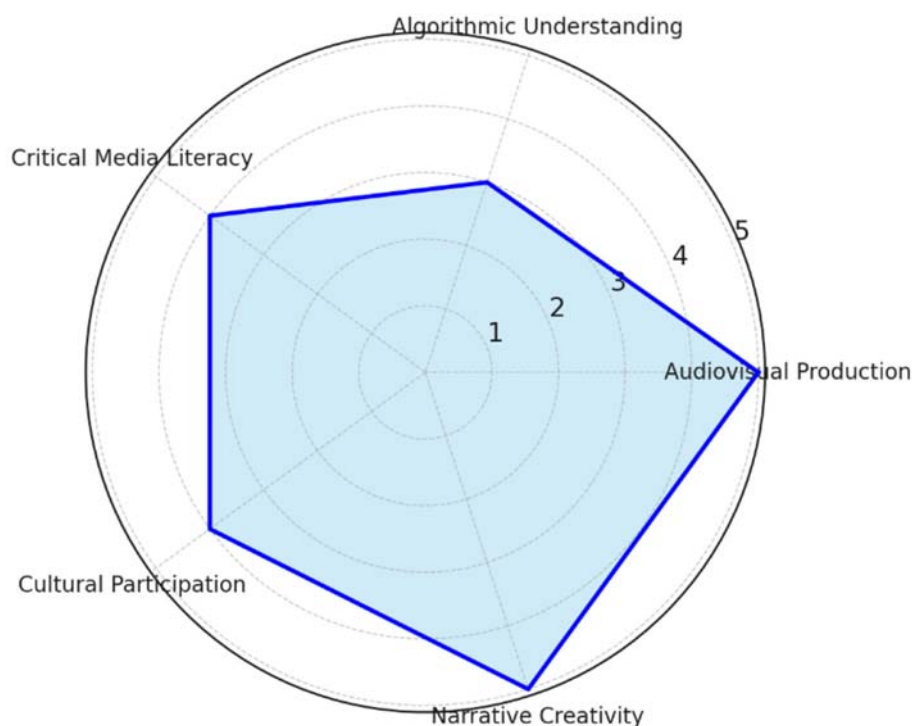


FIGURE 6
Digital and media literacies fostered by TikTok.

Mujica et al. (2024) and Núñez-Cook and Lizana (2024). At secondary level, studies focus on media- and information-literacy, emotional intelligence and academic performance, profiling how adolescents navigate the platform's information ecosystem rather than using it as a direct instructional channel. University-level research shifts to discipline-specific competence building, from international-business analytics to medical micro-video quality assessment and environmental-responsibility campaigns. Teacher-training studies interrogate digital self-efficacy and the emergence of "teacher-influencer" identities, revealing both pedagogical gains and tensions surrounding professional visibility. Finally, informal "BookTok" investigations demonstrate how vernacular, peer-led curation rekindles reading habits and reshapes local publishing ecosystems.

Documented learning effects—and the reporting gap. Quantitative evidence remains uneven. Of the 26 studies, only 15 use inferential statistics and merely one reports complete pre/post descriptive data, yielding a very large speaking-accuracy effect ($d = 2.13$).

The rest offer significance levels without means or standard deviations, limiting cumulative aggregation (see Table 4). This mirrors broader critiques of TikTok scholarship: recent systematic reviews on health literacy and higher-education usage likewise highlight promising outcomes but lament inconsistent statistical transparency.

Several authors explicitly situate their interventions within COVID-19 remote-learning mandates, arguing that TikTok's low entry barrier and affective design mitigated isolation and sustained learner engagement. Mixed-method classroom studies combine

TikTok production tasks with focus groups or reflective diaries, while analytics-based papers mine platform API data to triangulate behavioral insights—indicative of a methodological shift toward multimodal, data-rich designs.

Despite encouraging signals, the literature still debates whether TikTok is a transformative pedagogy or merely a motivational veneer. Empirical gaps persist around (i) long-term learning retention, (ii) the cognitive load imposed by rapid-fire content, and (iii) ethical dilemmas such as commercial influence on minors. Addressing these questions will require more robust experimental protocols.

3.3.1 Pedagogical and didactic approaches leveraging TikTok

Integrating TikTok into educational settings is reshaping pedagogical practice across all schooling levels, foregrounding creativity, collaboration, and learner agency. The 26 studies reviewed depict TikTok not merely as a content-delivery channel but as a participatory studio where students produce, remix and critique knowledge in ways that resonate with their digital lifeworlds.

One of the most pervasive designs is challenge-based learning via "duet", "stitch" and branded hashtag chains. In primary classrooms, 10-week micro-reading challenges (Diaz Mujica et al., 2024) and 81-s anatomy corrections (Núñez-Cook and Lizana, 2024) translate abstract literacy and terminology targets into playful, embodied tasks. Allemand and Bonfim (2021) extend this logic to dance education, showing how duet-based choreographic

prompts foster peer dialogue around movement vocabulary, while BookTok studies reveal similar dynamics as adolescents stage creative “shelf tours” and parody skits that revive print culture (Saez, 2022; Cuestas et al., 2022; Rovira Collado et al., 2024).

At the university level, TikTok underpins digital didactics aligned with meaningful-learning and active-constructivist ideals. International-business cohorts visualize market trends through 60-s explainer clips, yielding significant—albeit under-reported—performance gains for TikTok-literate students (Pozzo et al., 2024). Ecuadorian preservice teachers build research and linguistic competencies by scripting disciplinary mini-tutorials (Benitez-Correa and Gonzalez-Torres, 2025), while Peruvian engineering undergraduates co-create sustainability reels that boost environmental-responsibility scores (Romero Carrión et al., 2024). In biomedical fields, Izquierdo-Condo et al. (2025) demonstrate the platform’s reach—average 1.6 million views per clip—even as they caution that popularity metrics seldom align with JAMA quality benchmarks.

Teacher-training and professional-identity work form a second higher-education strand. A national survey of 213 Chilean journalism lecturers uncovers uneven self-perceived competence, with TikTok ranked the weakest platform (Lazcano-Peña et al., 2024). Ethnographic analyses of Brazilian “teacher-influencers” show educators commodifying lesson tips into performative content that reconfigures notions of pedagogical authority (Azzari and Mayer, 2022). Such studies underscore a double agenda: upgrading digital-production skills while negotiating the ethics of visibility and branding.

Across secondary contexts, TikTok serves primarily as a lens on youth information diets. Surveys in Colombia and Paraguay find the app competing with Facebook as a news source yet reveal limited correlation between usage time and academic performance (Paredes Peña et al., 2024; Sánchez Bernal et al., 2023). Feijoo et al. (2023) add a critical-advertising perspective, showing that Chilean tweens tolerate branded content only when it mimics the platform’s humoristic vernacular—raising questions about advertising literacy in algorithmic feeds.

These cases illustrate TikTok’s pedagogical elasticity: from micro-videos that visualize complex theories, through student-generated storytelling that reframes canonical texts, to thematic campaigns linking classroom concepts with civic or ecological discourse. Challenges remain—uneven reporting of learning gains, potential distraction, and data-privacy concerns—but the evidence points to substantial promise for didactic innovation when the platform is embedded within intentional, reflexive instructional designs.

3.3.2 Development of digital and media literacies

Beyond its direct instructional uses, TikTok is proving to be an effective catalyst for digital and media literacy (DML)—the ability to access, analyze, evaluate and create multimodal content in networked environments (Tiernan et al., 2023). Because every TikTok clip demands the orchestration of sound, text, movement and algorithmic timing, the platform naturally positions learners as producers rather than passive consumers, thereby activating the full cycle of DML practices.

In secondary and higher-education contexts, BookTok studies show students appropriating audiovisual rhetoric to perform literary criticism, blending close reading with trending audio and kinetic typography to reach wider publics (Botelho, 2024; Carvalho Rodrigues and Zenha Cordeiro, 2024). Similar multimodal “gossip-style” summaries prompted ninth-graders to negotiate genre conventions and authorial stance while navigating TikTok’s affordances (Botelho, 2024). These activities move literary analysis from the essay format into participatory fan-discourses, expanding students’ semiotic repertoire and audience awareness.

Survey research in the Colombian Amazon reveals that more than two-thirds of rural adolescent’s rank TikTok among their top news sources—yet also express low trust in platform information, signaling a need for explicit verification strategies (Paredes Peña et al., 2024). Parallel findings from Paraguayan high-schoolers show no direct performance penalty for heavy TikTok use, but underscore wide variability in critical-consumption habits (Sánchez Bernal et al., 2023). At university level, Izquierdo-Condo et al. (2025) correlate popularity metrics with JAMA scores for medical micro-videos and demonstrate that engagement is not a proxy for accuracy—an insight now incorporated into media-literacy workshops for health-science majors.

The technical demands of clip creation—layering captions, filters, transitions and duet/stitch sequences—cultivate hands-on mastery of audiovisual semiotics (Allemand and Bonfim, 2021). Engineering students producing climate-action reels reported significant growth in editing fluency and an improved ability to translate scientific evidence for lay audiences (Romero Carrión et al., 2024). Journalism faculties, however, reveal a skills gap: 35% of Chilean lecturers feel unprepared to generate TikTok content, with women and older staff reporting the lowest self-efficacy (Lazcano-Peña et al., 2024). This asymmetry highlights the importance of instructor DML upskilling to model critical practice.

Teacher-influencer ethnographies expose how curricular tips are re-packaged as entertainment, raising questions about labor, branding and authenticity in the attention economy (Azzari and Mayer, 2022). Children’s interviews in Santiago show they readily decode overt advertising cues but struggle with covert influencer marketing, valuing humorous and musical formats yet distrusting repetitive or deceptive spots (Feijoo et al., 2023). Such findings advance a nuanced literacy agenda: learners must grasp not only content construction but also the commercial and algorithmic forces that govern its circulation.

Collective knowledge building. Country-wide mappings of educational “EDUTOKERS” reveal that maths, tech and finance dominate Peru’s and Ecuador’s influencer landscape, furnishing peer exemplars that students cite as credible information gateways (González-Carrión et al., 2023). Cross-national Gen-Z surveys further show humor and self-image videos eclipsing news or academic clips, indicating that pedagogical interventions must insert critical-media tasks into genres already valued by young users (Martínez-Estrella et al., 2023).

The reviewed literature positions TikTok as a living laboratory for DML, where learners practice multimodal authorship, interrogate the reliability of user-generated content, and critique the algorithmic and commercial logics that structure digital discourse. When educators scaffold these practices—through guided production challenges, reflective analytics and explicit

source-evaluation heuristics—TikTok can nurture the transversal literacies essential for scholarly success and democratic participation in a data-driven society.

3.3.3 TikTok as a platform for identity construction and expression

TikTok's hybrid architecture—equal parts editing suite, social feed and reputation engine—invites learners to perform, remix and narrate their lived experience in ways that blur private storytelling with public display. Across the reviewed corpus, this expressive affordance emerges as a catalyst for identity work, allowing users to position themselves as readers, dancers, citizens, professionals or activists within algorithmically networked publics.

Reader-influencer identities like BookTok research illustrates how adolescents craft “micro-celebrity” personae centered on literary taste. Argentine BookTok creators deploy cinematic cuts, humor and spoiler-free reviews to claim cultural capital and peer authority, effectively translating close reading into platform vernacular. These performances extend beyond comprehension checks: they are acts of social positioning that renegotiate who counts as a legitimate literary critic in youth culture (Cuestas et al., 2022; Rovira Collado et al., 2024; Guíñez-Cabrera and Mansilla-Obando, 2022).

In Brazilian public-school dance projects, duet and remix functions enable children to co-create choreographies that assert agency and belonging, transforming standard technique drills into socio-aesthetic dialogues with classmates and remote partners (Allemand and Bonfim, 2021). Likewise, Afro-Colombian women featured in Truth-Commission micro-videos employ multimodal storytelling to weave individual experience into a collective memory of gendered violence, foregrounding intersectional identities of race, gender and regional marginalization (Rovira Collado et al., 2024).

A survey of 491 girls in low-SES schools across Colombia and Peru shows TikTok facilitating self-expression around emotional intelligence and human values; platform choice (Instagram vs. TikTok) correlates with subtle differences in perceived emotional clarity, suggesting that social-media ecologies mediate affective self-concept (Hoyos-Cifuentes et al., 2024). These findings echo Paraguayan data indicating that female students invest more time on TikTok yet outperform male peers academically, complicating deficit narratives around “screen time”.

At the opposite end of the educational pipeline, TikTok is reconfiguring what it means to be a teacher (Tiernan et al., 2023). Ethnographic analyses of “teacher-influencers” (Azevedo, 2023) reveal educators curating relatable personas that blend curricular tips with lifestyle vignettes, thereby commodifying pedagogical authority and negotiating authenticity in the attention economy. A profile-oriented scan of Peruvian and Ecuadorian “EDUTOKERS” (Salazar-Vallejo and Rivera-Rogel, 2024; González-Carrión et al., 2023) further shows STEM specialists accruing larger followings than humanities peers, signaling emergent hierarchies of disciplinary prestige on the platform. Yet a national survey of Chilean journalism lecturers exposes a contrasting identity gap: 35 % report low self-efficacy in TikTok content creation, with women and older staff most affected.

Youth self-image and cross-cultural variation. A four-country Gen-Z study finds humor, fashion and self-presentation videos dominating feeds, while language use emerges as the main marker of cross-national difference (Cervi, 2021; Martínez-Estrella et al., 2023; Saxena and Mishra, 2021). Such data underscore how TikTok's algorithmic curation simultaneously homogenizes global trends and amplifies local cultural codes, offering fertile ground for comparative identity research.

Commercial and ethical dimensions. Interviews with Chilean children show they decode overt advertising cues but resist covert influencer marketing, revealing an early awareness of the politics of visibility and persuasion on the platform (Feijoo et al., 2023). This critical stance aligns with teacher-education projects that position identity work as a pedagogical objective: preservice educators in Ecuador use TikTok to interrogate social-justice themes, balancing humor with epistemic rigor while reflecting on their own digital footprints (González-Carrión et al., 2023).

The reviewed studies confirm that TikTok is more than a content-sharing app; it is a dynamic arena of identity negotiation where learners and educators script who they are—and wish to become—within broader sociocultural and algorithmic frameworks. Harnessing this potential responsibly requires pedagogical scaffolds that encourage creative self-expression while cultivating critical awareness of the commercial, gendered and disciplinary logics that shape visibility online.

3.3.4 Emotional, cognitive and motivational dimensions of learning with TikTok

TikTok's appeal lies not only in its communicative reach but also in its capacity to activate intertwined affective, cognitive and motivational processes that underpin learning (Rininggayuh et al., 2024). The platform's bite-sized, audiovisual format leverages music, humor, rhythm and first-person narration to elicit swift emotional responses, which in turn scaffold attention and memory encoding—principles long recognized in multimedia-learning theory (Krüger and Bodemer, 2022).

Affective engagement and wellbeing. Classroom interventions consistently report high enjoyment and reduced anxiety when students migrate academic tasks to TikTok (Chao et al., 2023). Primary pupils exposed to 10 weeks of micro-reading clips showed both significant literacy gains and self-reported enthusiasm for “acting out” phonics on camera. In biomedical terminology, a single 81-s video normalizing the term *ulna* generated ≥ 82 % positive usefulness ratings, underlining the affective pull of concise, relatable visuals. Among vulnerable groups, the survey of 491 low-SES girls in Colombia and Peru found that producing values-themed TikTok's led to measurable improvements in emotional-intelligence clarity and repair dimensions, suggesting a therapeutic function of creative self-expression.

Cognitive processing and knowledge retention. TikTok's requirement to condense complex ideas into <60-s narratives encourage generative learning strategies such as summarizing, dual-coding and self-explanation. Speaking-accuracy research with Chilean undergraduates documented a very large effect ($d = 2.13$) after six picture-description microtasks, attributing gains to repeated retrieval and multimodal rehearsal. Engineering

students who storyboarded sustainability concepts into short clips likewise achieved statistically significant post-test improvements in environmental-responsibility scales—evidence that conceptual understanding can travel through performative formats. Yet analytics of 39 Spanish-language medical micro-videos warn that virality does not guarantee informational quality, underscoring the need for critical-evaluation scaffolds alongside cognitive offloading.

Motivational drivers and social reinforcement. TikTok is inherently gamified: trends, hashtags, likes and algorithmic visibility confer instantaneous feedback that activates both intrinsic interest and extrinsic recognition loops. University participants in the EDUTOKERS project reported heightened task value and audience awareness when assignments culminated in public-facing clips, aligning with self-determination theory's emphasis on autonomy and relatedness. Peer-evaluation mechanisms further amplify motivation; national data from Chile show that lecturers who feel more competent with TikTok perceive stronger student engagement and collaborative climate in their courses. Conversely, cross-sectional findings from Paraguay reveal no straightforward link between time on TikTok and grades, reminding educators that motivation must be channeled through purposeful design rather than sheer screen exposure.

Collectively, the evidence portrays TikTok as a platform capable of orchestrating emotion, cognition and motivation into mutually reinforcing cycles of learning. Emotion-laden micro-stories capture attention; condensed multimodal explanations support chunking and retrieval; and public, shareable outputs cultivate sustained interest and self-efficacy. However, the same features that drive engagement—brevity, virality, algorithmic reward—also risk superficial processing and unequal visibility. Future research should therefore integrate controlled comparisons of cognitive load, longitudinal tracking of retention, and rigorous quality-assurance protocols to harness TikTok's affective and motivational affordances without compromising deep learning.

3.3.5 TikTok and educational inequalities

While TikTok presents valuable pedagogical opportunities, its incorporation into educational contexts reveals persistent and emerging inequalities that demand critical scrutiny. These disparities span technological access, media representation, and socio-cultural capital, raising questions about who benefits from digital platforms in learning environments and under what conditions.

One of the most pressing challenges is the digital divide (Sam et al., 2025), which affects the availability of devices and connectivity and the capacity for productive and critical use of platforms like TikTok. Sam et al. (2025) note that students from rural or marginalized regions often face infrastructural limitations that prevent equitable participation in media-rich learning activities. These disparities become more pronounced when TikTok is used as a core pedagogical tool without parallel investments in digital literacy or access equity.

In addition to access, content dynamics within TikTok may reproduce dominant cultural norms, potentially reinforcing harmful stereotypes, aesthetic hierarchies, and exclusionary representations. Studies such as Saez (2022) and Allemand and Bonfim (2021) highlight that while students are often creative in

subverting norms, the broader platform ecology tends to favor mainstream and marketable identities, which can marginalize expressions rooted in indigenous, queer, or non-normative cultures. This risk is particularly salient in educational settings, where media use must align with inclusion, pluralism, and critical consciousness values.

Disinformation and algorithmic biases also pose ethical risks. TikTok's algorithm privileges engagement over accuracy, meaning educational content may compete with entertainment or misinformation, especially when learners are not trained to evaluate sources or discern quality. Wielki (2020) raised concerns over the role of influencers and visual manipulation in shaping perceptions among youth, highlighting the need for critical media education as a countermeasure to informational inequalities.

Despite these challenges, several initiatives demonstrate the potential of TikTok to function as a tool for educational justice. The EDUTOKERS project (Salazar-Vallejo and Rivera-Rogel, 2024) and the study by Izquierdo-Condoy et al. (2025) showcase how students from disadvantaged backgrounds can appropriate TikTok as a space for agency, creativity, and civic expression when guided by reflective pedagogies. These projects exemplify a shift toward media empowerment, where educators intentionally scaffold digital tools to amplify marginalized voices and democratize learning.

Therefore, the use of TikTok in education must be framed as a matter of innovation, equity, and ethics. This involves designing interventions that mitigate digital disparities, foreground inclusive narratives, and equip learners with the tools to interrogate and reshape digital culture from within critically.

3.4 Ethical considerations

The integration of TikTok into educational research and practice raises ethical questions that extend beyond traditional classroom settings. As a proprietary commercial platform governed by opaque algorithms, TikTok shapes what content becomes visible, how users are profiled, and how interactions circulate publicly. These dynamics influence educational participation and must be acknowledged when evaluating its pedagogical potential. Several of the studies included in this review relied on publicly available TikTok content or student-generated videos, yet ethical reporting varied substantially across publications. In educational interventions involving minors or vulnerable groups, clear procedures regarding informed consent, privacy protection, and data anonymization are essential, particularly when videos remain accessible beyond the scope of the study.

Algorithmic bias constitutes a further ethical challenge. TikTok's recommendation system may amplify certain linguistic, cultural, or aesthetic norms while suppressing others, potentially reinforcing existing social disparities. For South American learners—especially those from rural, Indigenous, or low-income communities—algorithmic visibility can shape whether their contributions are recognized or marginalized. Researchers must therefore consider how platform logics intersect with regional inequalities and avoid interpreting algorithmically shaped behaviors as student characteristics or learning outcomes.

The use of public data in educational research also requires careful scrutiny. Although TikTok videos may be openly accessible,

participants do not always anticipate that their content will be analyzed for academic purposes. Ethical international guidelines recommend obtaining explicit permission whenever identifying information is present or when the content involves minors. Researchers working with scrape-based datasets must ensure compliance with institutional review boards and regional data-protection standards, such as Brazil's LGPD, Argentina's Habeas Data laws, and evolving Andean Community regulations.

From a practical standpoint, ethical integration of TikTok in educational settings demands more than compliance. Educators should prepare students to critically interpret algorithmic recommendations, understand how their digital traces are stored and circulated, and reflect on the implications of producing content in commercial platforms. Schools and universities must also ensure equitable access to devices and connectivity to prevent exacerbating digital divides. These considerations highlight that the educational promise of TikTok is inseparable from the ethical infrastructures that govern its use, requiring institutions, researchers, and educators to adopt transparent, context-aware and rights-based approaches to platform-mediated learning.

4 Discussion

The purpose of this review was to consolidate empirical and conceptual knowledge regarding the educational uses of TikTok in South America, a region characterized by profound sociocultural diversity and persistent structural inequalities. The analysis of the included studies reveals an expanding yet methodologically heterogeneous field, where TikTok functions simultaneously as a pedagogical tool, a cultural artifact, and a space for youth agency. While the platform's multimodal and participatory affordances appear to align with emerging pedagogical paradigms in the region, the evidence base remains uneven, requiring careful interpretation.

One of the most salient findings concerns the ways in which TikTok-mediated practices intersect with educational inequality. Several studies indicate that the platform may serve as an alternative entry point into learning for students who experience barriers in traditional school environments. Through short-form video creation, algorithmically mediated visibility, and culturally resonant formats, learners from rural, low-income, or otherwise marginalized backgrounds gain opportunities to participate in forms of expression, collaboration, and meaning-making that may be less accessible in conventional academic settings. These dynamics suggest that TikTok can facilitate more horizontal modes of mediation between teachers and students, echoing the dialogic principles proposed by Freire and subsequent South American educommunication scholars. However, this potential is not uniformly realized. Unequal access to devices, connectivity, digital competencies, and institutional support continues to shape who can meaningfully participate in TikTok-based educational activities. The coexistence of democratizing affordances and reproduction of inequities positions the platform as both a tool with transformative promise and an object that reflects broader socio-digital asymmetries.

The meta-analytic and quantitative synthesis contributes further nuance to this interpretation. Although only one study provided sufficient descriptive detail to compute an effect size,

that study suggested substantial gains associated with TikTok-mediated speaking activities. The remaining studies, assessed through structured vote counting, generally reported positive associations between TikTok use and improvements in motivation, emotional engagement, environmental responsibility, scientific communication, or literacy practices. Nonetheless, methodological limitations—including small sample sizes, lack of control groups, reliance on self-report, absence of pre/post measures, and incomplete statistical reporting—limit the strength of the claims that can be made. The diversity of designs underscores the need for more rigorous, transparent, and repeatable empirical approaches to studying digital platforms in the region.

Risk-of-bias implications for interpretation. The methodological limitations noted above are also reflected in the ROBINS-I risk-of-bias assessment (Table 1) for the quantitative evidence base. Overall risk of bias ranged from moderate to serious, with confounding and deviations from intended interventions as the most recurrent domains driving these ratings. In practical terms, this means that many reported “positive effects” may partially capture pre-existing differences between groups (e.g., baseline proficiency, prior digital competence, motivation, teacher support, and socio-economic context) or uncontrolled variation in exposure, engagement, and co-interventions rather than TikTok-specific causal impacts. Consequently, the quantitative findings should be read as indicative associations that support the plausibility of educational benefits, but not as definitive estimates of effect magnitude. This is particularly important given the limited extractability of effect sizes and the reliance on vote counting for most outcomes; therefore, conclusions about effectiveness remain provisional until future South American studies adopt stronger designs (e.g., controlled or matched comparisons with baseline adjustment, preregistered outcomes, standardized measures, and complete reporting) that reduce bias and enable confirmatory meta-analytic synthesis.

Recent international evidence positions TikTok as a rapidly consolidating educational microlearning environment across disciplines, but with effects that depend strongly on pedagogical design and on the platform's engagement affordances. A recent systematic review of TikTok in higher education reports expanding disciplinary adoption and generally positive learning-related outcomes (e.g., engagement, perceived usefulness, participation), while also highlighting persistent implementation barriers (e.g., superficial engagement, ethical concerns, classroom governance) and a continued shortage of rigorous evaluative designs that would allow stronger causal claims (Yang et al., 2025). Empirical work in higher education similarly suggests that TikTok can support microlearning and student acceptance when content is intentionally structured and aligned with course goals (Conde-Caballero et al., 2024), and that students' active learning intention is shaped by perceived usefulness/ease of use and platform-specific affordances such as interactivity, immersion, playfulness, and virtual rewards (Wang et al., 2024). In parallel, scholarship has documented how “TeachTok” practices reconfigure teacher identity and classroom authority through micro-celebrification and participatory “fun learning” communities (Vizcaino-Verdú and Abidin, 2023), a dynamic that is consistent with the relational shifts and horizontal mediation patterns identified in our South American synthesis.

Comparatively, our findings converge with this broader body of knowledge in two core ways: first, the predominance of affective engagement and learner agency as central mechanisms through which TikTok mediates participation and motivation; and second, the recurring need for explicit scaffolding to prevent entertainment logics and algorithmic visibility dynamics from displacing curricular depth. At the same time, our review contributes a region-specific positioning by showing that, in South America, TikTok's educational potential is more tightly entangled with structural inequality and unequal digital conditions than is typically foregrounded in the wider higher-education literature. This contextual emphasis complements prior evidence that educational use of TikTok is conditioned by platform affordances and user motivations (Rahimullah et al., 2022), while also aligning with findings that educational TikTok ecosystems may privilege humor and attention capture over deliberative debate, thereby raising challenges for critical pedagogy if instructional framing is weak (Tejedor Calvo et al., 2022). In short, the international evidence supports the plausibility of benefits under well-designed microlearning integration, but our South American synthesis highlights that effectiveness and equity are inseparable: future work in the region will need stronger designs and fuller reporting to estimate effects credibly, while also explicitly measuring access conditions, digital competencies, and institutional supports that shape who can benefit and how.

A notable thematic pattern involves the centrality of student identity construction and affective engagement. Studies on language learning, scientific outreach, and digital storytelling show that TikTok encourages students to assume active roles as content creators, critics, and cultural agents. This aligns with theories of participatory culture and mediated learning, suggesting that TikTok fosters environments in which learners negotiate identities, collaborate with peers, and visualize cognitive processes through creative expression. However, the review also highlights that such opportunities depend heavily on educators' pedagogical framing. When activities are carefully scaffolded, students can deepen their reflective capacities and engage more critically with media structures. When scaffolding is minimal or absent, learning becomes susceptible to algorithmic bias, superficial engagement, or displacement of academic objectives by entertainment logics.

The platform's relationship with edcommunication traditions in South America also merits attention. Several studies emphasize the transformative role of TikTok in reconfiguring the teacher-student relationship toward more horizontal, collaborative practices. Through co-creation, dialogic feedback loops, and aesthetic experimentation, TikTok can act as a mediator that destabilizes hierarchical classroom structures and enhances the sense of proximity and reciprocity between educators and learners. At the same time, such relational shifts demand professional development, clear ethical guidelines, and institutional recognition to avoid placing undue burdens on teachers or reinforcing hidden curricular expectations tied to digital labor.

Despite promising indications of TikTok's pedagogical relevance, the review exposes a series of structural and epistemic challenges that must be addressed. Foremost among these is the persistence of the digital divide, which manifests not only in infrastructure disparities but also in differences in digital literacy, teacher preparedness, and institutional

capacity. Urban centralization of technological resources further widens educational gaps, especially in countries such as Peru, Ecuador, and Colombia, where rural regions experience chronic underinvestment. Algorithmic opacity adds another layer of complexity, as platform dynamics influence what students see, how they are positioned as content creators, and how their work circulates. These issues are particularly pronounced for marginalized youth whose cultural productions may be rendered invisible or misinterpreted by platform algorithms.

Because each quantitative synthesis included a limited number of studies, we did not assess reporting bias and did not rate certainty of evidence; therefore, pooled estimates should be interpreted cautiously.

Future research emerging from the present synthesis should therefore attend not only to learning outcomes but also to the epistemic nature of the content produced and consumed on TikTok. The review highlights a lack of studies examining the quality, accuracy, and ideological framing of educational content, as well as the sociotechnical mechanisms through which algorithms amplify or suppress particular forms of knowledge. Addressing these gaps is essential for understanding how TikTok participates in broader struggles over representation, voice, and educational justice in South America.

The findings suggest that TikTok holds meaningful but uneven potential for pedagogical innovation in the region. Its multimodal affordances, cultural proximity to youth, and possibilities for expressive learning align with contemporary visions of inclusive and participatory education. Yet, realizing this potential requires intentional pedagogical design, equitable access policies, ethical safeguards, and sustained institutional support. As the evidence base evolves, researchers and practitioners must adopt critical, context-sensitive approaches that foreground both the affordances and constraints of platform-mediated education in South America.

5 Conclusions

The evidence reviewed positions TikTok as a multifaceted learning environment that can amplify motivation, deepen multimodal literacy and give voice to under-represented learners across South America. Yet empirical work is concentrated in a handful of countries and higher-education settings, leaving primary, indigenous and community contexts largely unexplored. Positive outcomes arise when TikTok activities are framed by clear learning objectives, critical-media scaffolds and opportunities for peer feedback. In the absence of such design principles, the same affordances that energize students—brevity, virality and algorithmic reward—risk superficial processing or inequitable visibility. Connectivity constraints, limited device access and uneven staff training continue to delimit who can create, share and benefit from TikTok-based tasks. Addressing these structural barriers is a pre-condition for scaling platform-mediated innovations beyond well-resourced urban schools and universities. Future studies should:

- Adopt longitudinal and experimental designs that report full descriptive and effect-size statistics;

- Extend inquiry to marginalized linguistic, rural and Afro-descendant communities;
- Examine the cognitive load and long-term retention associated with ultra-short video learning;
- Interrogate the ethical implications of datafication, commercial branding and algorithmic bias in classroom use.

When coupled with equity-oriented pedagogy and robust digital-literacy frameworks, TikTok can serve as more than a conduit for curricular content: it can become a participatory arena where students rehearse civic voice, negotiate identity and co-construct knowledge. Realizing these potential demands interdisciplinary collaboration among educators, researchers, technologists and policy-makers committed to shaping socially just, evidence-based digital education.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

AR-C: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing. MC-G: Conceptualization, Data curation, Methodology, Software, Writing – review & editing. SC-M: Data curation, Formal analysis, Investigation, Writing – original draft. PN-B: Conceptualization, Methodology, Project administration, Supervision, Writing – review & editing.

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Supplementary material

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SUPPLEMENTARY MATERIAL S1
PRISMA 2020 checklist (completed).

SUPPLEMENTARY MATERIAL S2
Full-text exclusion table ($n = 142$): complete list of excluded full-text articles with the primary exclusion reason.

SUPPLEMENTARY MATERIAL S3
Data extraction spreadsheet for quantitative synthesis and effect-size calculations.

SUPPLEMENTARY MATERIAL S4
Variable coding codebook describing all extracted variables and coding rules.

SUPPLEMENTARY MATERIAL S5
R script used for effect-size calculations and meta-analysis.

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